

Rock Mechanics

Introduction

Engineering activities involving rock have been underway for millennia



By BoneA - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=51945782>

Ġgantija ("**Giantess**") is a megalithic temple complex from the Neolithic on the Mediterranean island of Gozo.

- earliest of the Megalithic Temples of Malta
- older than the pyramids of Egypt.
- two Ġgantija temples during the Neolithic (c. 3600–2500 BC)
- more than 5500 years old and the world's second oldest existing manmade religious structures after Göbekli Tepe in present-day

There are more than 1,500 known rock-cut structures in India. Many of these structures contain artwork of global importance, and most are adorned with exquisite stone carvings.



The Barabar Hill Caves are the oldest surviving [rock-cut caves in India](#), dating from the [Maurya Empire](#) (322–185 BCE), located in the [Makhdumpur](#) region of [Jehanabad district](#), [Bihar](#), [India](#), 24 km (15 mi) north of [Gaya](#).

The monolithic [Kailash Temple](#) is considered to be the peak of the Rock Cut Cave Type construction at the Ellora Caves, Maharashtra, India. A megalith carved from a rock cliff face, it is considered one of the most remarkable cave temples in the world because of its size, architecture and sculptural treatment.



By Prathepps - Own work, CC BY-SA 2.5, <https://commons.wikimedia.org/w/index.php?curid=794937>



Lead and zinc mines in Rajasthan, India, were examined, in relation to, especially, the early use of zinc as a metal. Carbon dates of circa 2000 years before present established Zawar as the world's oldest zinc mines. Large-scale mining for lead at Rajpura-Dariba goes back to circa 3000 years before present. (<https://www.jstor.org/stable/124574>)



<https://timesofindia.indiatimes.com/city/jaipur/Gossan-rich-monument-a-forgotten-age-old-site/articleshow/51861451.cms>

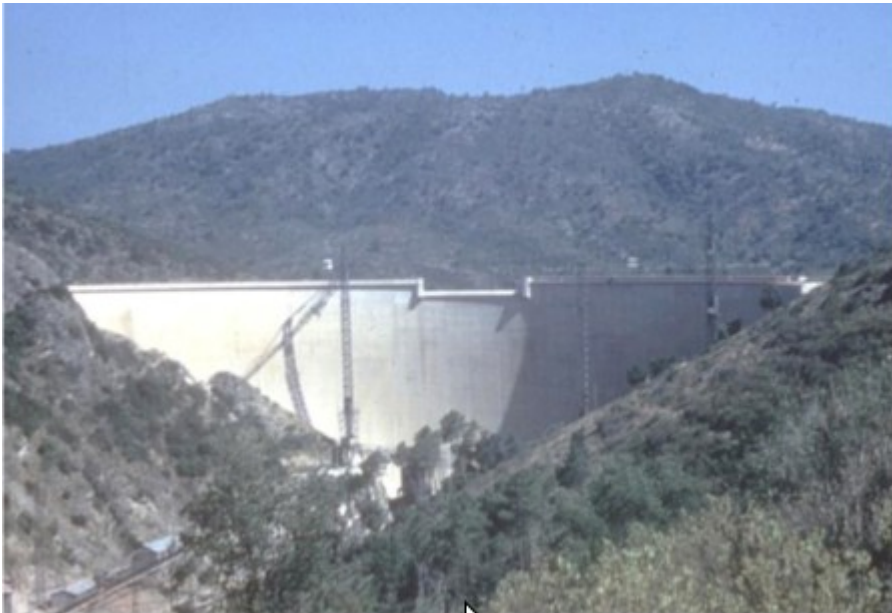
Notable Timelines

1773	Coulomb included results of tests on rocks from Bordeaux in a paper read before the French Academy in Paris (Coulomb, 1776, Heyman, 1972)
1884	▪ French engineers started construction of the Panama Canal. ▪ Taken over by the US Army Corps of Engineers in 1908. ▪ In the half century between 1910 and 1964, 60 slides were recorded in cuts along the canal. ▪ these slides were not analysed in rock mechanics terms at that point of time ▪ Lutton et al, (1979) showed that these slides were predominantly controlled by structural discontinuities and that modern rock mechanics concepts are fully applicable to the analysis of these failures. • Karl Terzaghi (Terzaghi, 1936, Terzaghi and Voight, 1979) said • <i>‘The catastrophic descent of the slopes of the deepest cut of the Panama Canal issued a warning that we were overstepping the limits of our ability to predict the consequences of our actions’.</i> • <i>in discussing the Panama Canal slides in his Presidential Address to the first international conference on Soil Mechanics and Foundation Engineering in 1936,</i>
1920	▪ Josef Stini started teaching ‘Technical Geology’ at the Vienna Technical University ▪ before he died in 1958 he had published 333 papers and books (Müller, 1979). ▪ He founded the journal Geologie und Bauwesen, the forerunner of today’s journal Rock Mechanics, ▪ <i>Josef Sitni was probably the first to emphasise the importance of structural discontinuities on the engineering behaviour of rock masses</i>

Von Karman (1911) King (1912) Griggs (1936) Ide (1936) and Terzaghi (1945)	Research of Failure of Rocks
1921	Griffith proposed his theory of brittle material failure
1931	Bucky started using a centrifuge to study the failure of mine models under simulated gravity loading.

None of the aforementioned persons would have classified themselves as rock engineers or rock mechanics engineers - the title had not been invented at that time - but all of them made significant contributions to the fundamental basis of the subject as we know it today.

The early 1960s were very important in the general development of rock engineering world-wide because *a number of catastrophic failures occurred which clearly demonstrated that, in rock as well as in soil, 'we were over-stepping the limits of our ability to predict the consequences of our actions' (Terzaghi and Voight, 1979).*



<http://dx.doi.org/10.1016/j.jrmge.2013.07.004>

Malpassee Dam

Location: Fréjus, France

Year Constructed: 1954

Type: Masonry/Concrete

Height: 218 ft.

Primary Purpose: Irrigation,
Water Supply

Date of Incident: December
2, 1959

Evacuation: No

Fatalities: 421

<https://damfailures.org/case-study/malpassee-dam-france-1959/>

Lessons Learnt From the Disaster

1. Concrete gravity dams should be evaluated to accommodate full uplift.
2. Dam incidents and failures can fundamentally be attributed to human factors.
3. Stability of the dam foundation and other geologic features must be considered during dam design.
4. Timely warning and rapid public response are critical to saving lives during a dam emergency.
5. External independent peer review of designs and decisions is an effective means of providing quality assurance and reducing the risk associated with design oversights and deficiencies.

Vajont Dam

Location: Italy

Year Constructed: 1957-1960

Type: Masonry/Concrete

Height: 860 ft.

Primary Purpose: Hydropower

Date of Incident: October 9, 1963

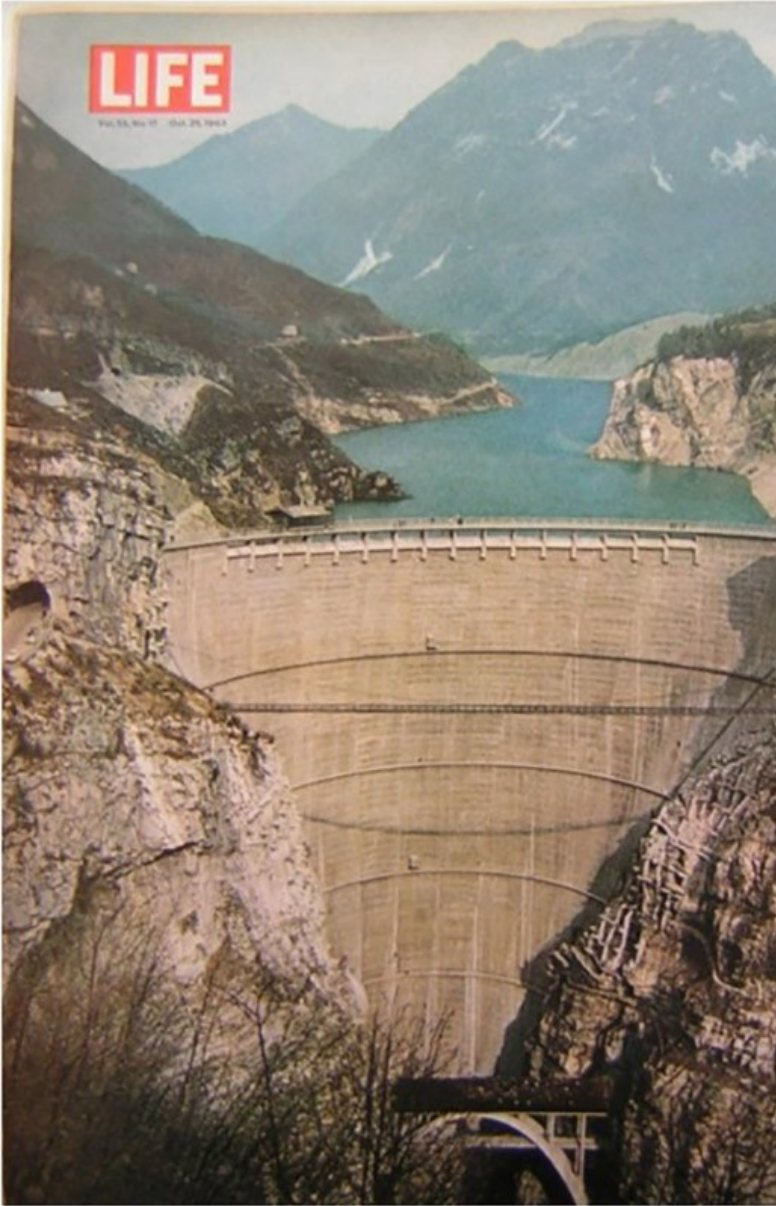
Evacuation: No

Fatalities: 2,056



<https://damfailures.org/case-study/vajont-dam-italy-1963/>

- the Vajont reservoir began filling in February 1960.
 - During first filling there were differing opinions on whether the reservoir bank along Mount Toc was the site of an ancient rockslide. *A local newspaper published an article suggesting that landslides were recurrent near the dam site and increased the likelihood of a disaster.* This created a reaction with the public and shortly after, legal action was taken against the newspaper for spreading false news and disturbing the peace. Residents were assured of safety, but they continued to distrust the stability of Mount Toc, nicknaming it “the mountain that walks”. Dam construction was completed in September 1960.
 - November 4, 1960 when, with the reservoir water level at a depth 600 feet (elev. 645 meters), a rockslide of almost **one million cubic yards** entered the reservoir. A seven foot wave propagated across the reservoir as a result of the landslide, but there were no injuries.
 - Subsequent draw downs and fillings allowed based on the response of creep monitoring in the basin.
 - Several days before the disaster, the mayor of Erto issued a manifesto urging villagers to evacuate. A few homes were evacuated, but there was little concern.
-
- The mayor of Casso ordered people to evacuate slopes and posted notice of an expected wave from an anticipated landslide. An hour and *an half before the disaster, traffic was restrained on roads below the dam, while telephone messages said “maybe a little water over the dam tonight, (but).. nothing to be alarmed about.”* The electric company was said to have told an engineer to “stay calm and sleep with his eyes open”.



On the night of October 9, 1963, a massive rockslide occurred with the reservoir 782 feet deep (elev. 700.4 meters). At the time of slide, the reservoir level was 82 feet below the dam crest elevation and the reservoir volume was around 109,000 acre-feet. At approximately 1.2 miles wide by 1.0 mile deep, the enormous mass was estimated at 350 million cubic yards of earth and debris. The volume of the slide was about twice the volume of water impounded behind Vajont Dam and slid into the reservoir in less than 45 seconds.

Vajont is considered the most disastrous rockslide ever to occur in Europe. The event created strong earth tremors, recorded as far away as Vienna and Brussels. The rockslide displaced the Vajont reservoir, causing wind and water to travel in all directions. An updraft of air, rocks, and water climbed the canyon walls to around 850 feet above the reservoir. Waves of water reached greater than 330 feet above the dam crest. Approximately 20,000 acre-feet of water went over the dam, creating a flood depth at the mouth of Vajont River of about 230 feet, one mile downstream.

Five downstream towns were destroyed including Longarone, Pirago, Rivalta, Villanova, and Fae. The water first arrived in Longarone at 10:43 pm, around 4 minutes after the rockslide. The town was completely leveled due to the depth and speed of the flood wave (high severity flooding). Of an estimated 1,328 people in Longarone who were at risk at the time of the floodwave, 1,269 fatalities occurred, resulting in a fatality rate of 94 percent. Longarone was a tourist destination, a mountain village made up of mostly residential and commercial structures. The flood wave completely destroyed the town, resulting in one of the highest known fatality rates ever recorded due to flooding from a dam failure or incident.

Surprisingly, the Vajont Dam structure sustained little damage, even though subjected to a force from the combined slide and overtopping wave far in excess of design pressures. The dam remains in place today, albeit with no water behind it. Though the dam structure appears to have been well designed (only the top several feet of the dam suffered damage), designers didn't understand the geology in the valley basin.

Lessons Learned

1. Dam failure sites offer an important opportunity for education and memorialization.
2. Dam incidents and failures can fundamentally be attributed to human factors.
3. Dams located in seismic areas should be evaluated for liquefaction, cracking, potential fault offsets, deformations, and settlement due to seismic loading.
4. Emergency Action Plans can save lives and must be updated, understood, and practiced regularly to be effective.
5. Intervention can stop or minimize consequences of a dam failure. Warning signs should not be ignored.
6. Stability of the dam foundation and other geologic features must be considered during dam design.
7. The first filling of a reservoir should be planned, controlled, and monitored.
8. Timely warning and rapid public response are critical to saving lives during a dam emergency.
9. Additional Lessons Learned
10. External independent peer review of designs and decisions is an effective means of providing quality assurance with design oversights and deficiencies.
11. Geo-hazards can exist at dams, even if a structure is considered safe and does not fail.
12. Thorough geotechnical investigations are needed during dam design, specifically reservoir slope stability analyses for dams in mountainous areas.
13. A clear, concise warning message with specific details are more likely to be heeded during an emergency situation.S

The 1960 Coalbrook Disaster

Date : 21 January 1960

Mine : Coalbrook, situated in the Northern Free State 21km south west of Vereeniging.

Lives lost : 437 miners

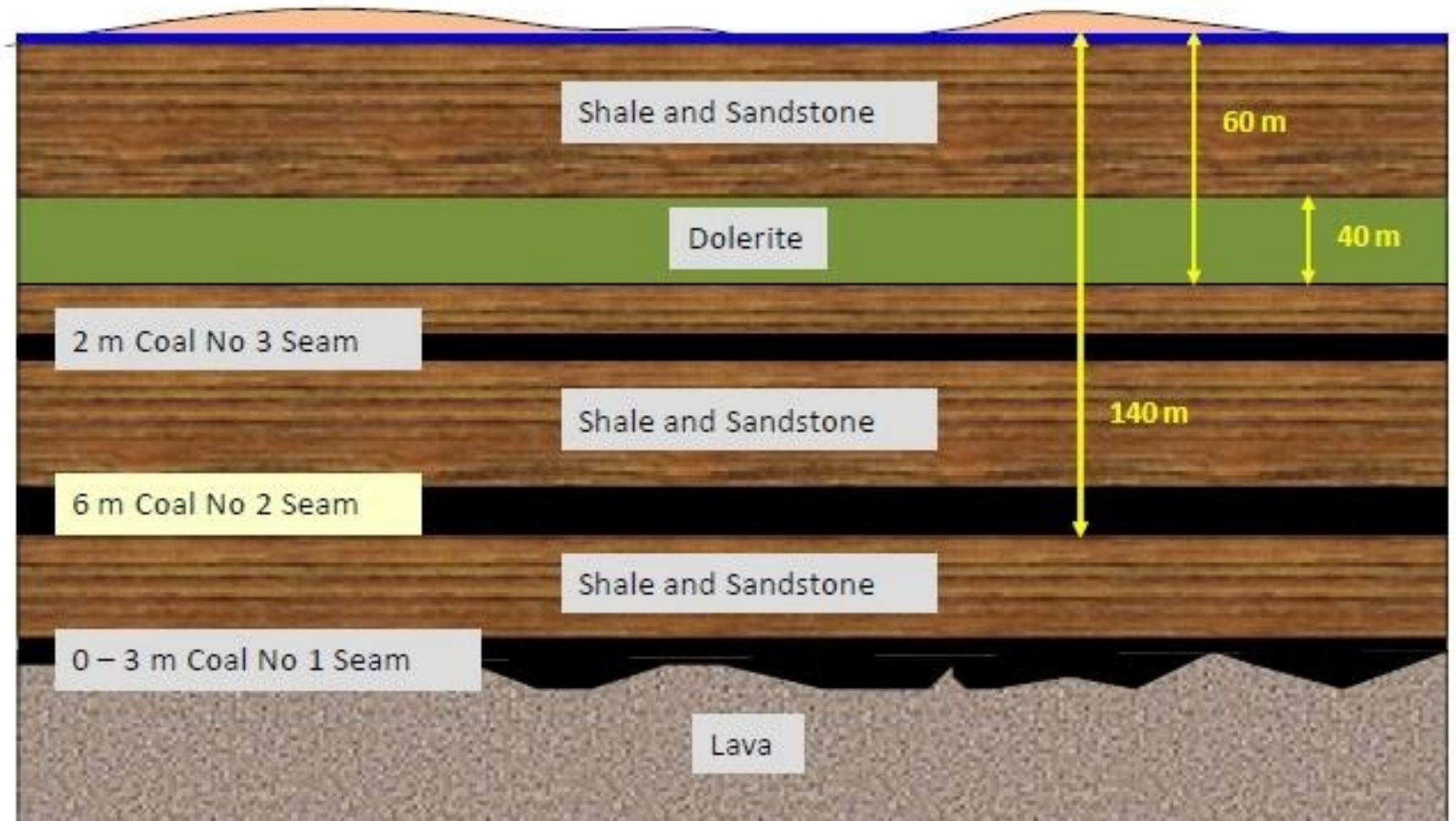
Depth of working : 180 meters below the surface

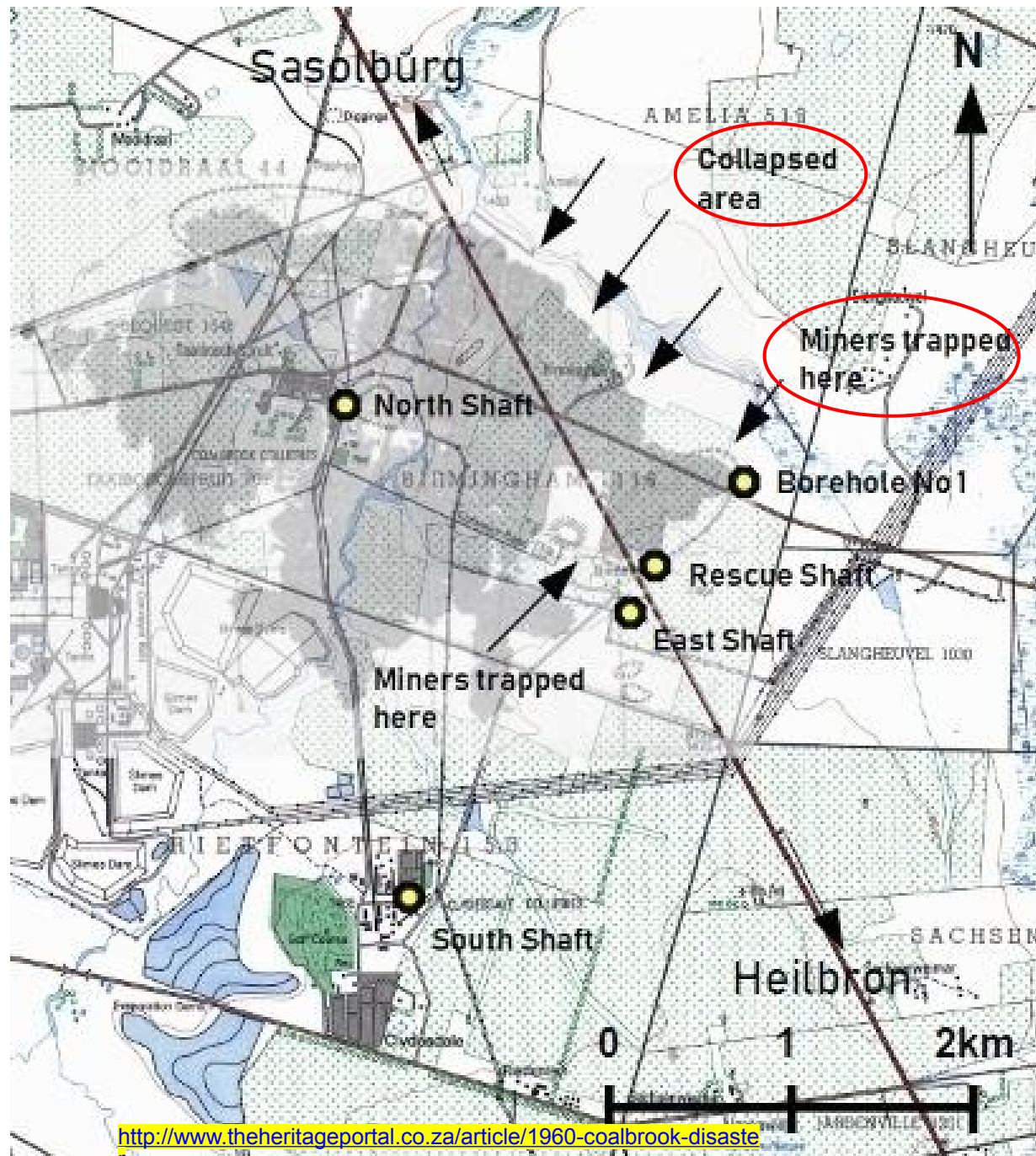
Number of Pillar collapse: estimated 900

- The coal seam (No 2) under active extraction. (6 m thick)
- This was only mined to 2.4m height.
- The material above that was of lower quality and of no interest to the market served at the time.
- In 1948 the mine had already been experimenting with extracting coal from higher up in the strata by raising the height to 3.7m.
- The term used for this was 'top coaling'. Top coaling was started again in 1951 to supply the new power station. Later the height was further increased to 4.5m.

This event was responsible for the initiation of an intensive research programme which resulted in major advances in the methods used for designing coal pillars (Salamon and Munro, 1967).

Geology and Dimensions





- On the 28 December 1959 a collapse occurred in the northern part of the mine (section 10), which included the area where the top coaling experiment was done.
- That collapse was arrested by a barrier wall to the south. Coal extraction from the south end of the section was not affected.
- An inspector of mines came on a routine inspection two weeks later. He was not informed about the collapse.
- At 4.00pm on 21 January 1960 miners in a section just west of the collapsed area heard shot-like noises and felt a strong blast wind.
- The area was evacuated and the mine manager informed. Miners from the area just south also heard the noises and withdrew.
- The acting mine manager and overseer entered the mine to inspect. Some of the ventilation piping had been damaged. At the same time it was reported that there was a subsidence on the surface of section 10.
- It was concluded that the 'weight had come off' and it was time to send in a repair team.
- Production in the south-eastern section was not affected and mining there carried on uninterrupted.
- The repair team working on the ventilation piping became aware of thundering noises at about 7:00 pm and withdrew to a safe area. On the way they were overtaken by a hurricane of dust and more noises. A general exodus from the mine ensued.
- It was later noticed that not a single miner from the south east side of section 10 had come out and it had to be assumed that they were trapped.

The beginning of the modern era of Rock Mechanics is marked from the year 1962 when the International Society for Rock Mechanics (ISRM) was formally established in Salzburg, Austria.

“Rock mechanics is the theoretical and applied science of the mechanical behavior of rock; it is that branch of mechanics concerned with the response of rock to the force fields of its physical environment”

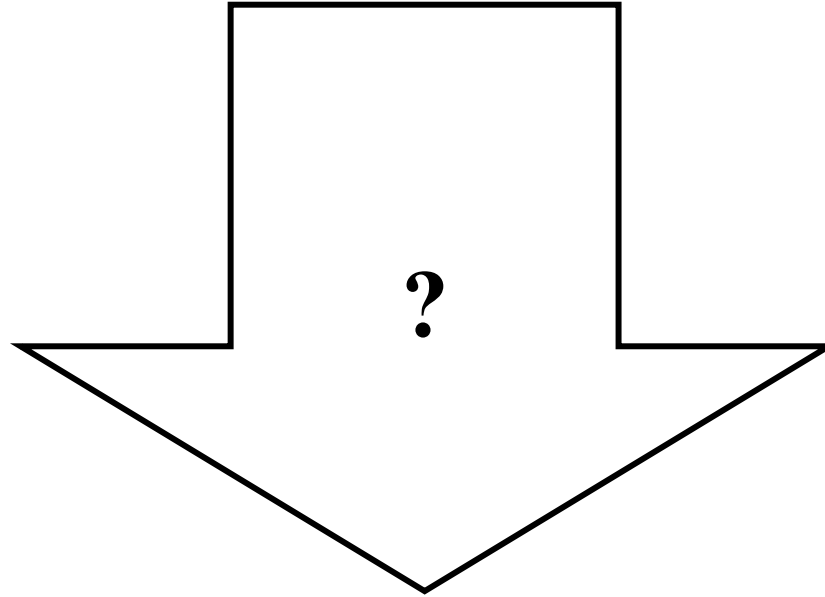
*(Judd, 1964).
Committee on Rock Mechanics of the
Geological Society of America*

The Classical Theory of SOM

Metals

Yield is Important

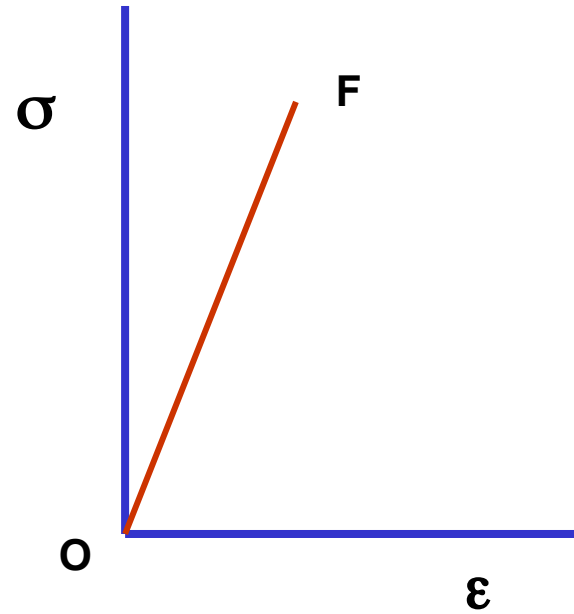
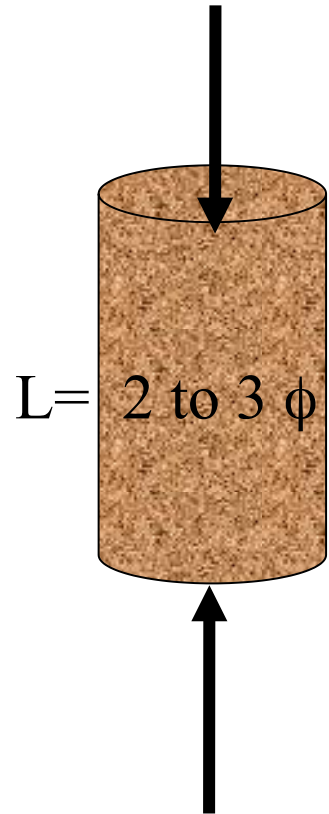
Difference in
Macroscopic
as well as
Microscopic Level



Failure & Fracture is important

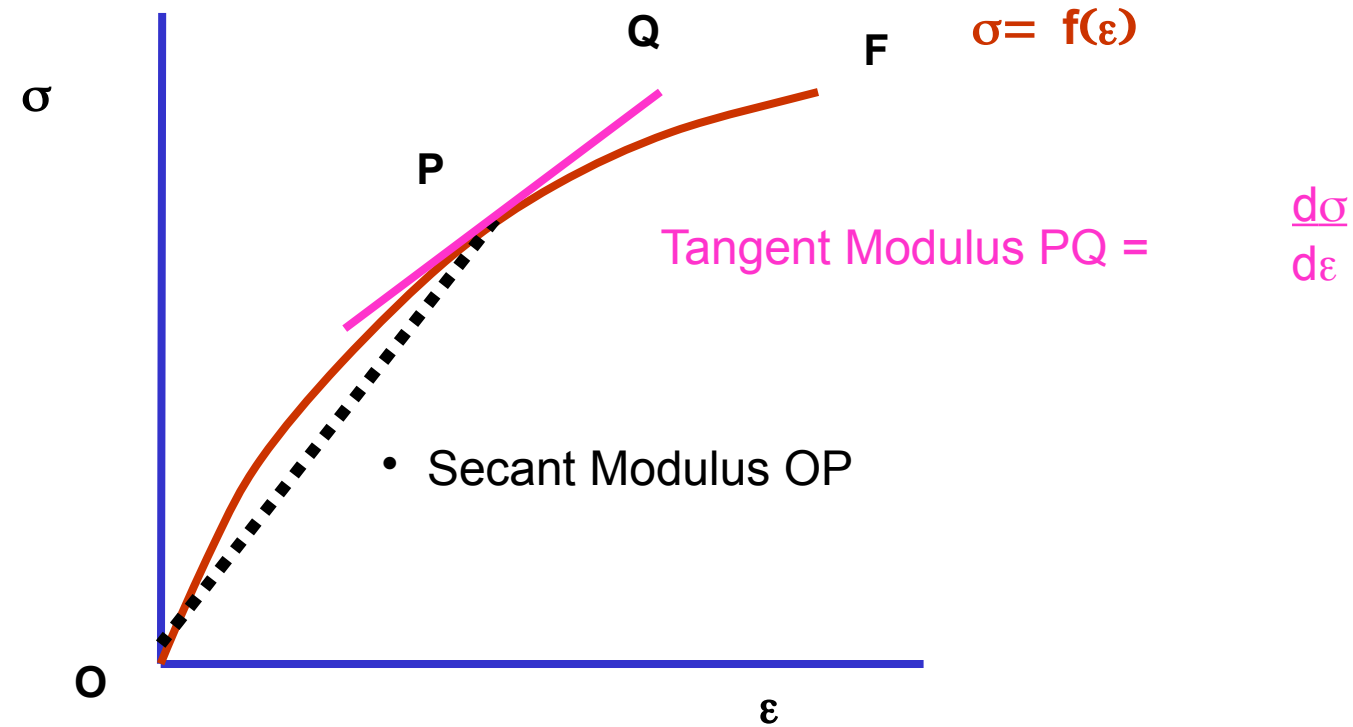
Rocks

Stress Strain Curve



Linearly Elastic Material.

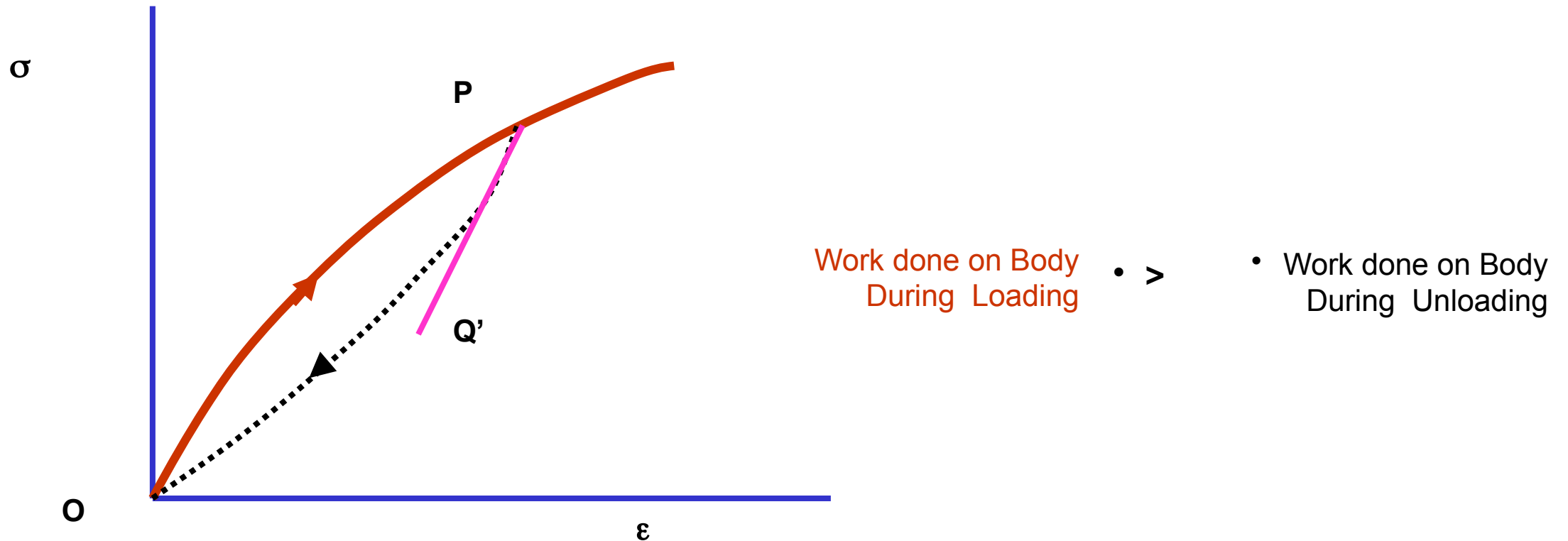
Perfectly Elastic Material



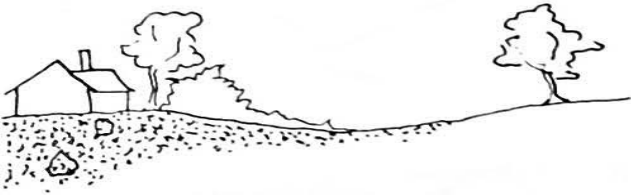
- Loading and Unloading Path are Same.

Elastic Material With Hysteresis

Showing Loading and Unloading Cycle.

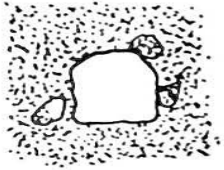


Conventional engineering design	Rock engineering design
• The first assessment is to determine the loads to be applied externally on the structure • Material is then prescribed with appropriate strength and deformation characteristics. (CHILE Material – Continuous, Homogeneous, Isotropic and Linear Elastic) • Then structural geometry is selected	• Designer has to deal with complex rock mass (DIANE Material - Discontinuous, Inhomogeneous, Anisotropic and Nonlinear Elastic) • Specific material cannot be prescribed to meet design requirements • Applied loads are not as important in rock masses as forces resulting from re-distribution of original stresses • Number of possible failure mode can exist in rock structures. Hence determination of material strength is a major problem • Geometry of structure may depend on configuration of geological structure.

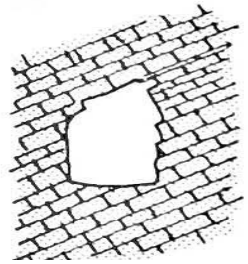


Stability problems encountered with increasing depth below surface

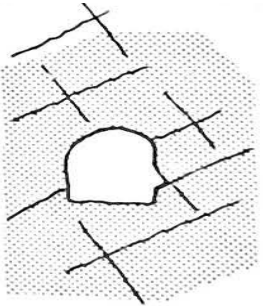
Increasing depth below surface



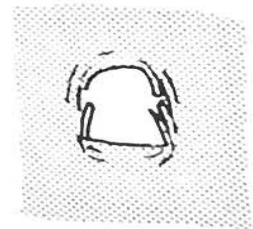
Overburden soil and heavily weathered rock – squeezing and flowing ground, short standup time



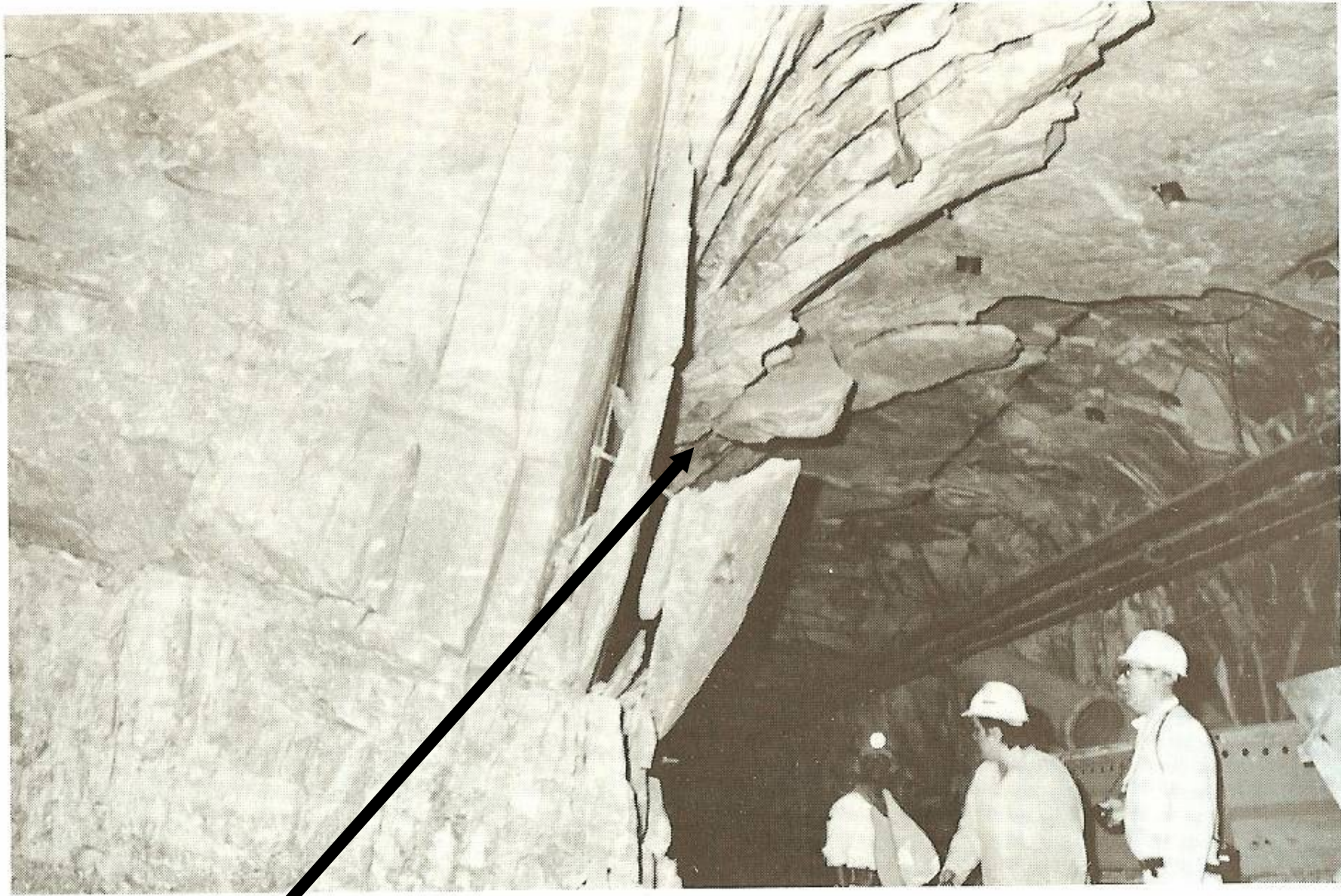
Blocky jointed rock partially weathered – gravity falls of blocks from roof and sidewalls



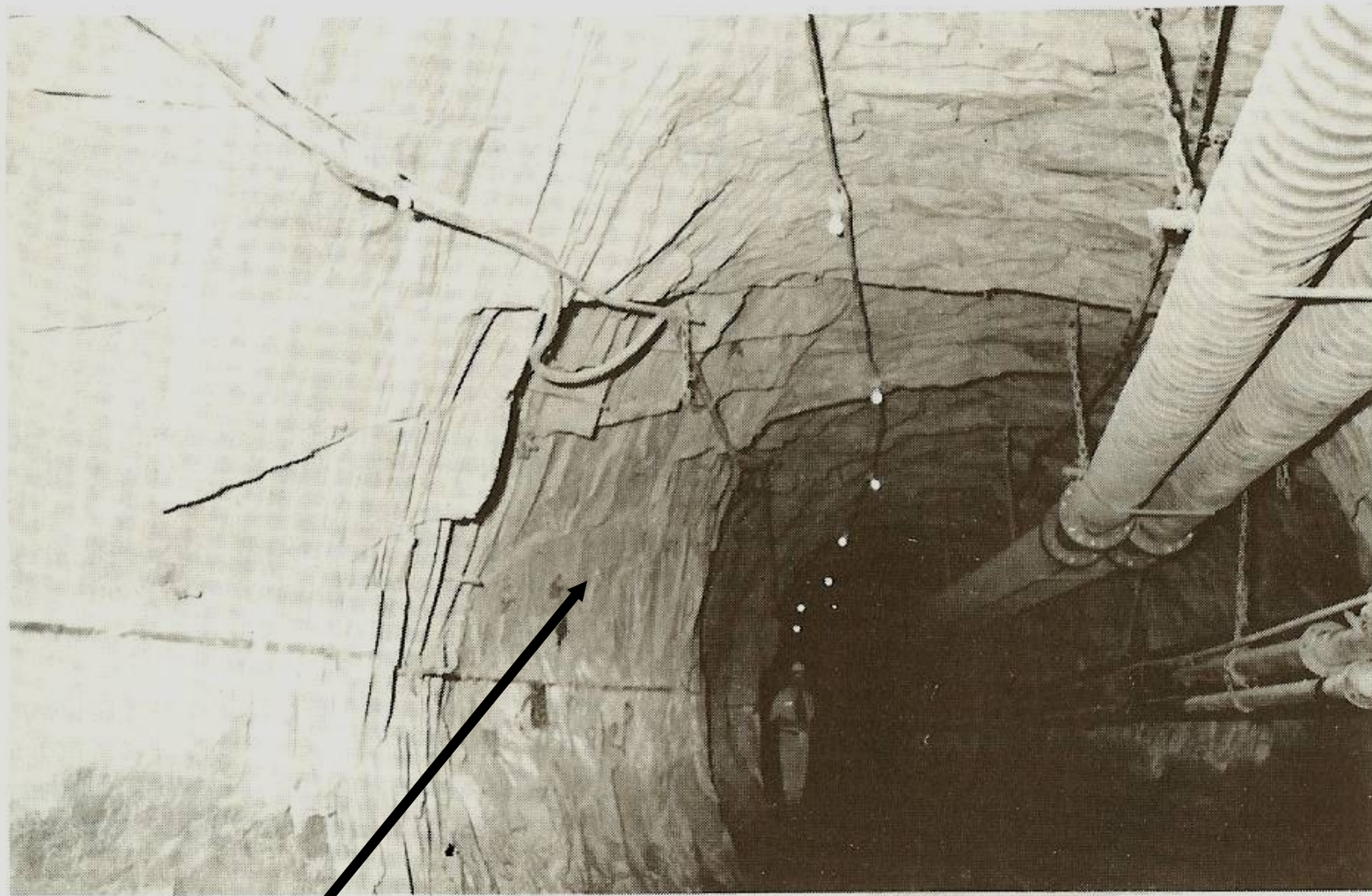
Massive rock with few unweathered joints – no serious stability problems



Massive rock at great depth – stress induced failures, spalling and popping with possible rockbursts



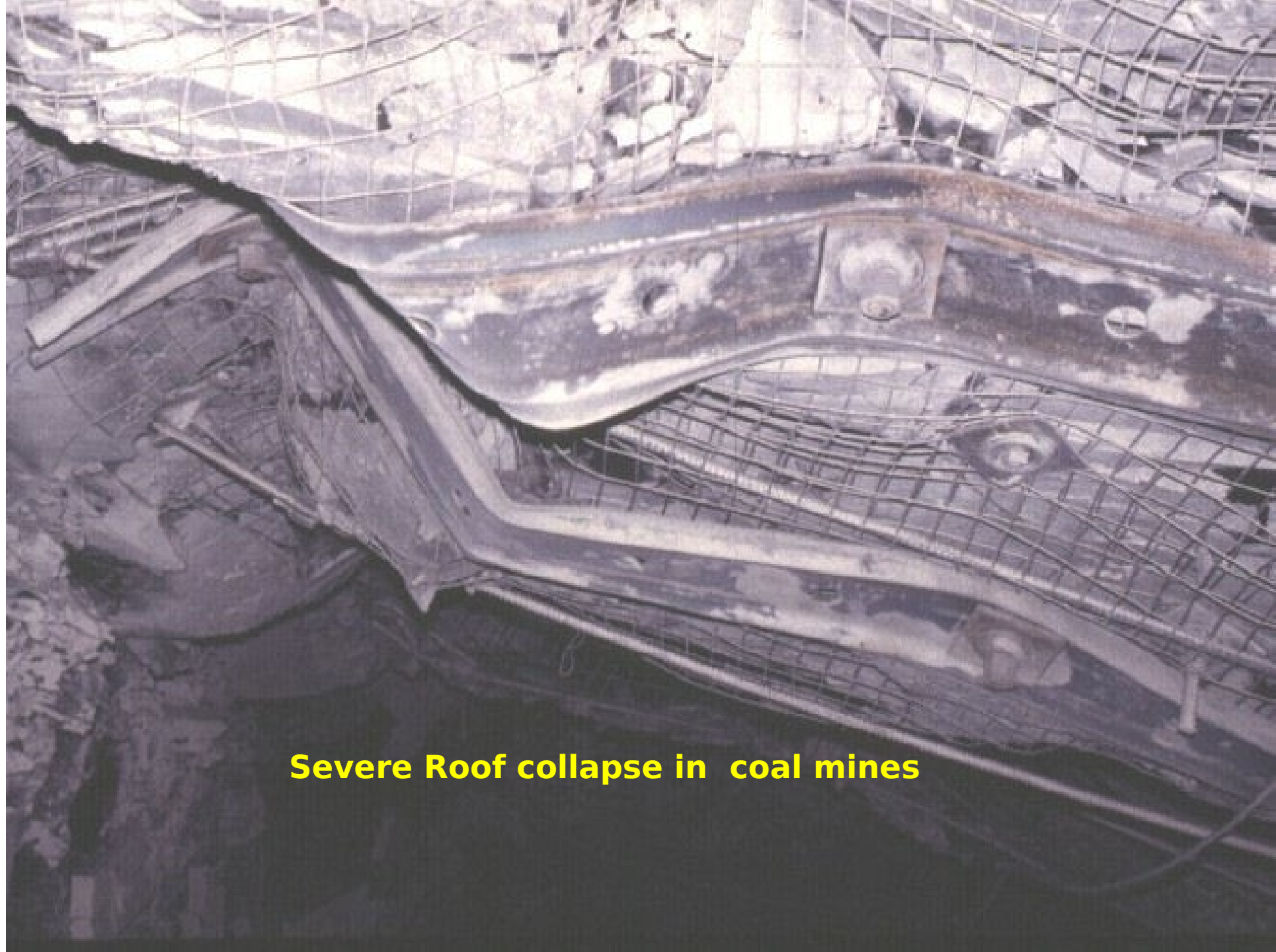
Side Wall Slabbing (Stress Induced failure)



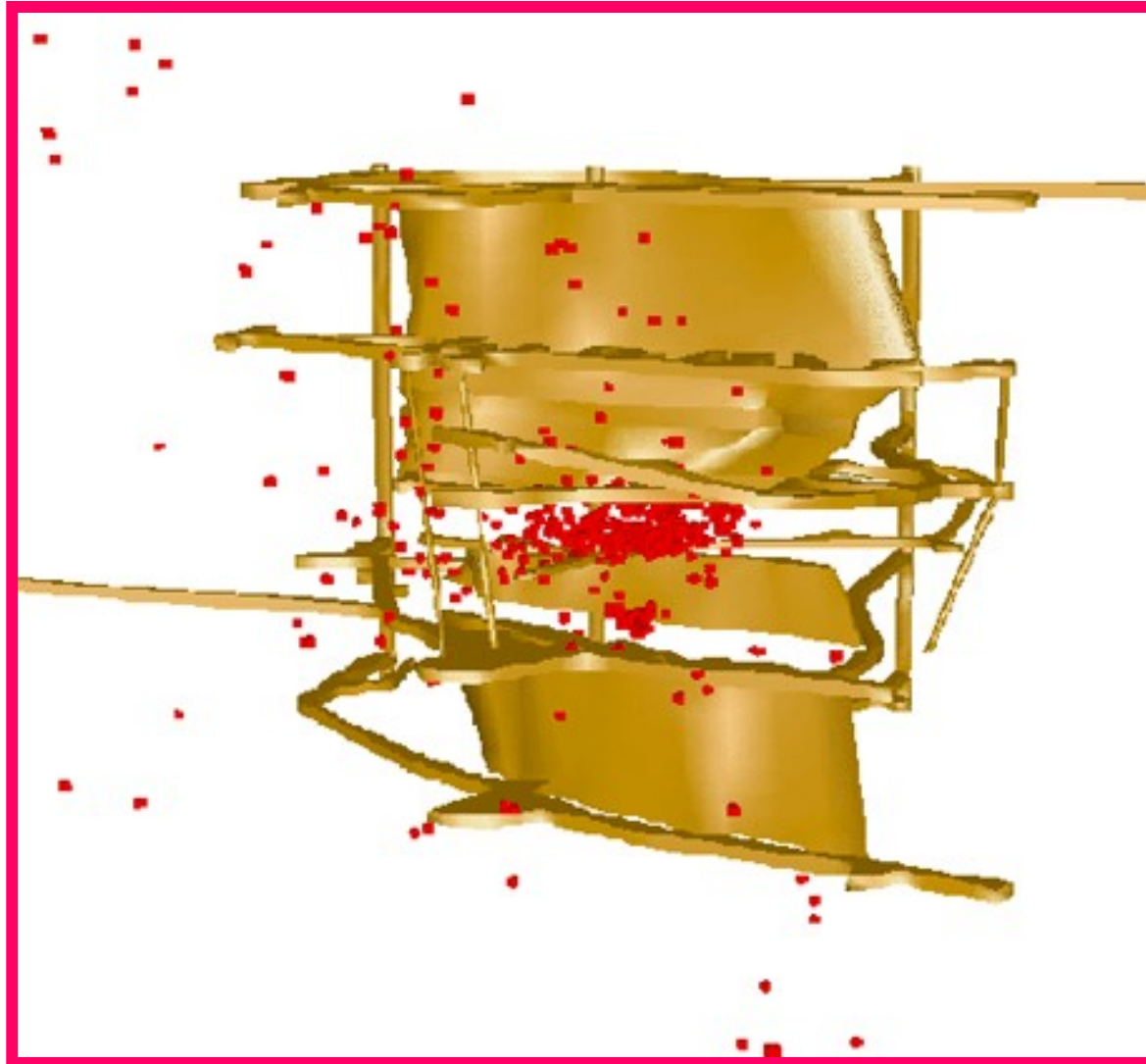
Concavity In The Sidewall (Indication Of High Stress)



**Collapse
of
sidewalls**



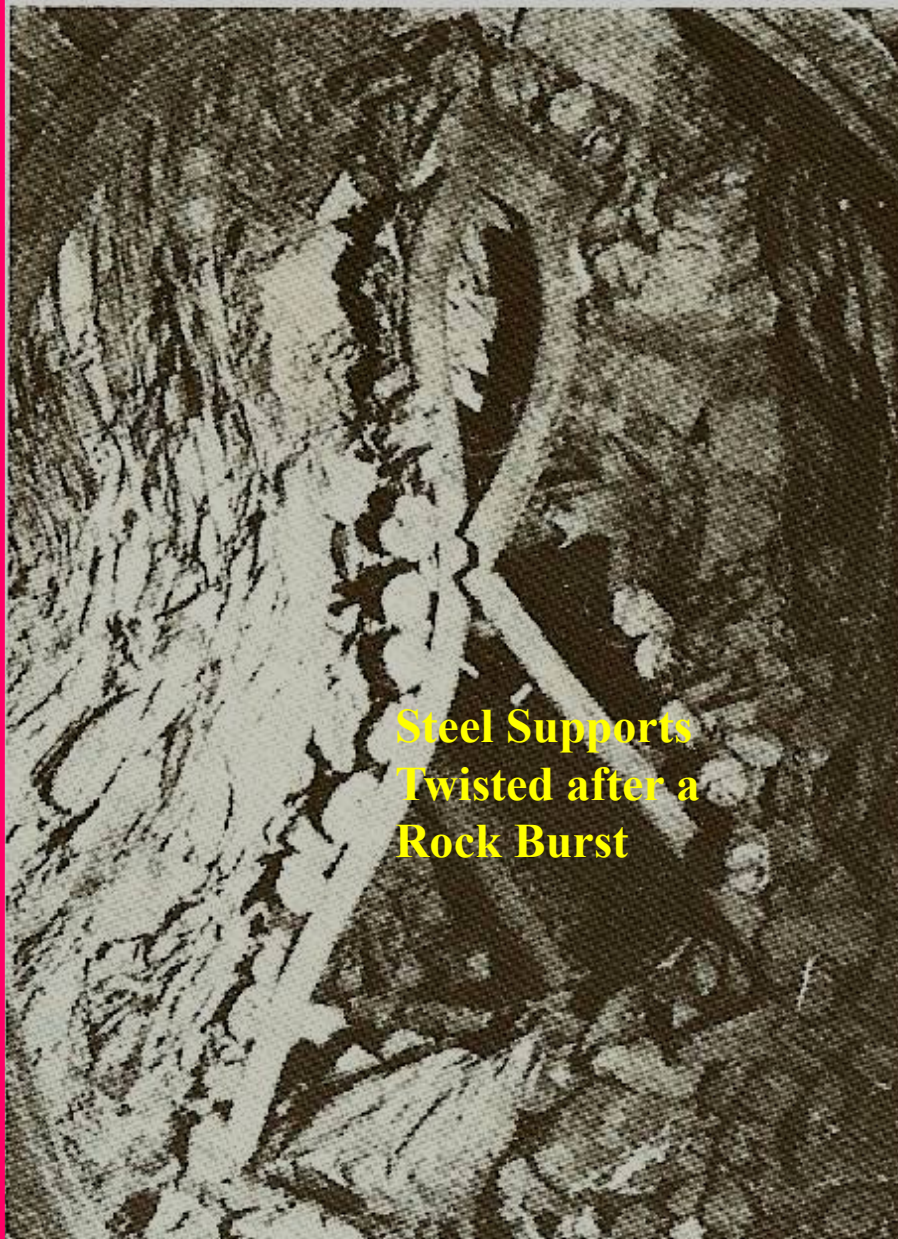
Severe Roof collapse in coal mines



Micro-seismic Event Locations at one of the Mines

A black and white photograph showing the aftermath of a severe rock burst in a tunnel. The scene is dark and chaotic, with large, jagged rock fragments scattered throughout. A wire mesh netting is visible on the left side, partially collapsed and covered in debris. Several wooden poles or beams are leaning against the rock walls, some appearing to be part of the tunnel's support structure. The overall atmosphere is one of destruction and danger.

Aftermath of a severe Rock Burst

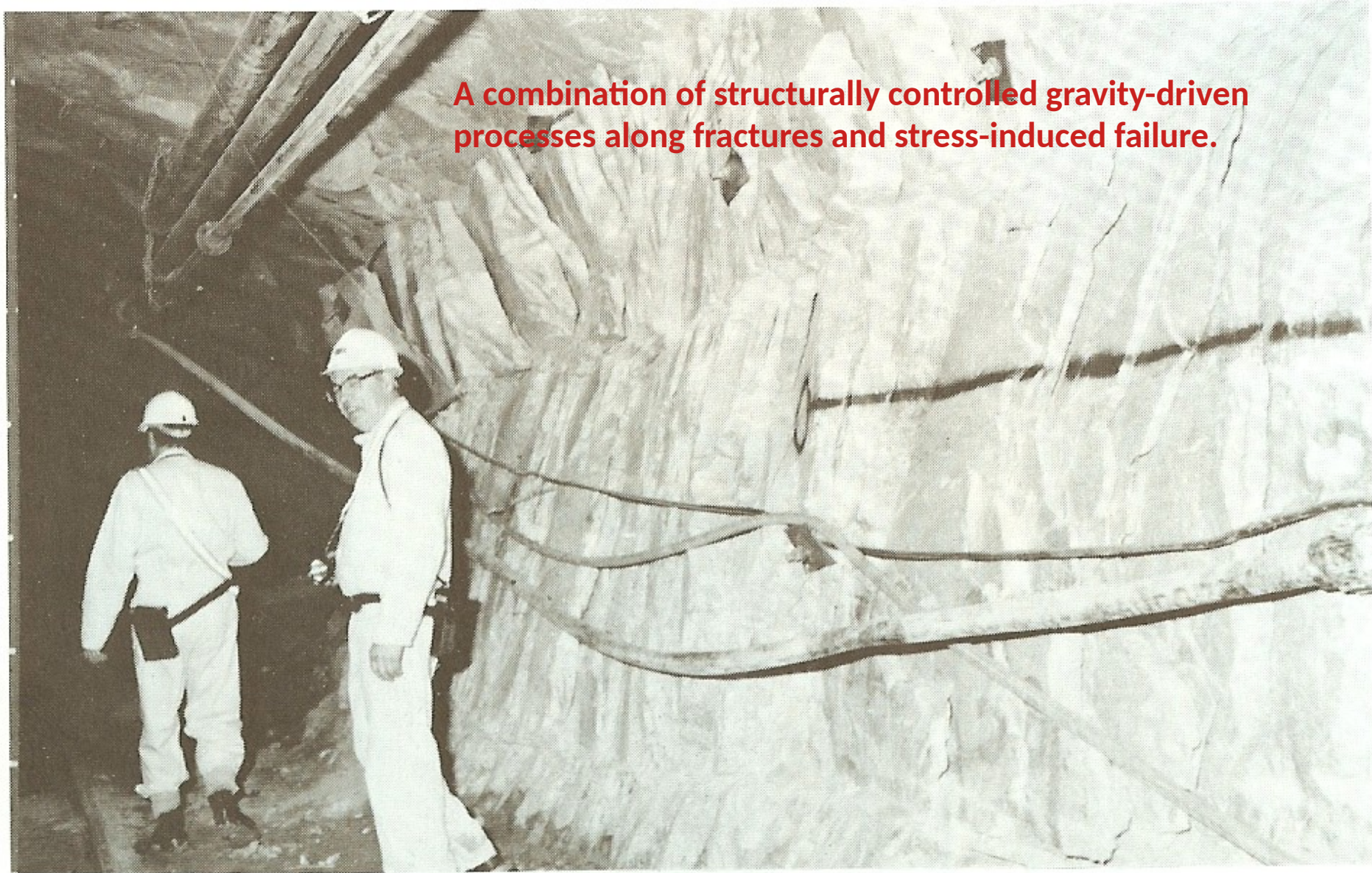


**Steel Supports
Twisted after a
Rock Burst**



**Original Shape
of the Steel
Supports**

Rock Burst Damage at Kolar Gold Field Mines



Intermixing of rock structural feature and fracture initiation due to stress @ Depth of 2700 m.

Design Method in Rock Engineering

```
graph TD; A[Design Method in Rock Engineering] --> B[Analytical Method]; A --> C[Observational Method]; A --> D[Empirical Method]; B --> B1[Analysis of stress around the excavation]; B --> B2[Techniques]; B2 --> B2a[Closed form solutions]; B2 --> B2b[FEM, BEM, FD, DEM etc]; B2 --> B2c[Analog Simulation (electrical & photoelastic)]; B2 --> B2d[Physical modelling]; C --> C1[Relies mainly on actual monitoring of ground movement during excavation]; C --> C2[Analysis of ground support interaction]; C2 --> C3[This approach is the only way to check results and predict the methods of design]; D --> D1[Statistical analysis of u/g excavations]; D1 --> D2[Engineering classification of rock mass is the best known approach for assessing the excavations in rock];
```

Analytical Method

- Analysis of stress around the excavation

Techniques

- Closed form solutions
- FEM, BEM, FD, DEM etc
- Analog Simulation (electrical & photoelastic)
- Physical modelling

Observational Method

- Relies mainly on actual monitoring of ground movement during excavation
- Analysis of ground support interaction

This approach is the only way to check results and predict the methods of design

Empirical Method

- Statistical analysis of u/g excavations

Engineering classification of rock mass is the best known approach for assessing the excavations in rock

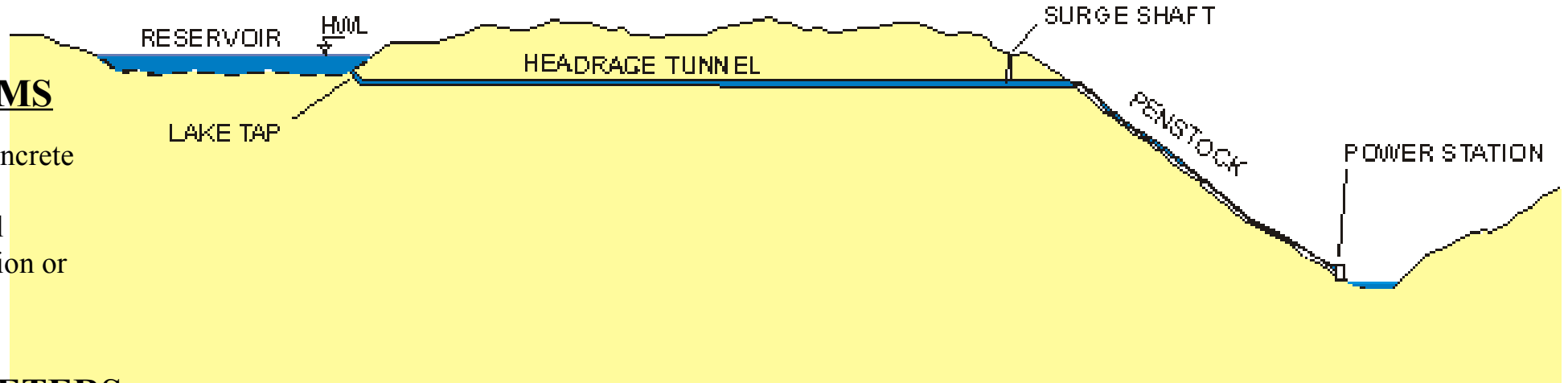
PRESSURE TUNNELS IN HYDROPOWER PROJECTS

1. TYPICAL PROBLEMS

- Leakage from unlined or concrete lined tunnels
- Rupture or buckling of steel lining due to rock deformation or external pressure

2. CRITICAL PARAMETERS

- Ratio of maximum hydraulic pressure in tunnel to minimum principal stress in surrounding rock
- Length of steel lining and effectiveness of grouting
- Ground water levels in rock mass



3. ANALYSIS METHODS

- Determination of minimum depth of cover along pressure tunnel from topographic maps.
- Stress analyses of sections along and across tunnel axis
- Comparison between minimum principal stresses and maximum dynamic hydraulic pressure to determine steel lining lengths

4. ACCEPTIBILITY CRITERIA

- Steel lining is required where **minimum principal stress in the rock is less than 1.3 times the maximum static head for typical hydroelectric operation** or 1.15 for operations with very low dynamic pressure.
- Hydraulic pressure testing in boreholes at the calculated ends of the steel lining is essential to check the design assumptions

1. TYPICAL PROBLEMS

- Rock failure where induced stress exceeds the strength of the rock.
- Swelling, squeezing or excessive closure if support is inadequate

2. CRITICAL PARAMETERS

- Strength of rock mass and of individual structural features
- Swelling potential, particularly of sedimentary rocks
- Excavation methods and sequence
- Capacity and installation sequence of support systems

3. ANALYSIS METHODS

- Stress analysis using numerical methods to determine extent of failure zones and probable displacements in rock mass
- Rock support interaction analysis using closed form or numerical methods to determine capacity and installation sequence for support and to estimate displacements in rock mass

4. ACCEPTIBILITY CRITERIA

- Capacity of installed support should be sufficient to stabilize the rock mass and to limit closure to an acceptable level.
- Tunneling machines and internal structures must be designed for closure of the tunnel as a result of swelling or time dependent deformation.
- Monitoring of deformation is an important aspect of construction control.



1. TYPICAL PROBLEMS

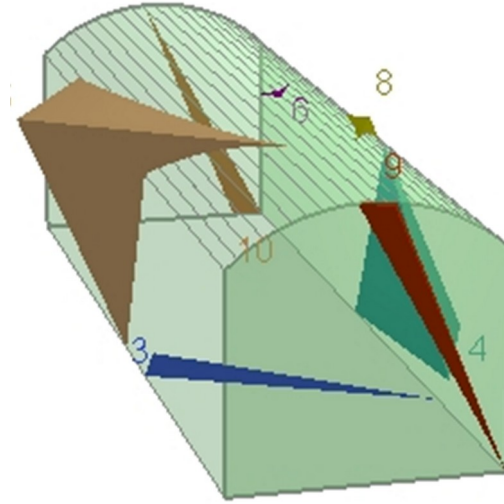
- Gravity driven falling or sliding wedges or blocks defined by intersecting structural features
- Unravelling of inadequately supported surface material

2. CRITICAL PARAMETERS

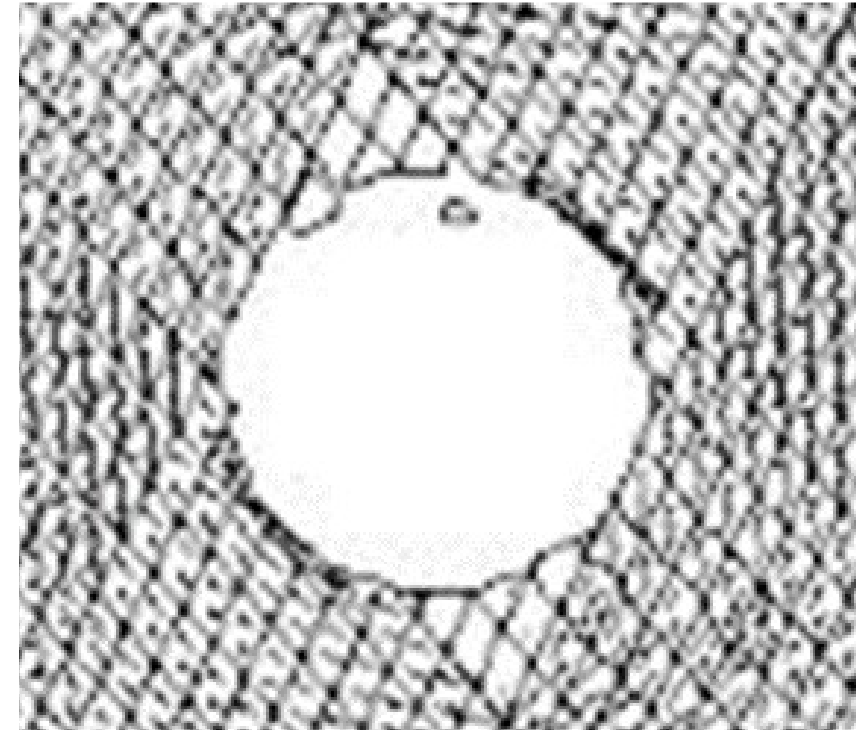
- Orientation inclination and shear strength of structural features in rock mass
- Shape and orientation of excavation
- Quality of drilling and blasting during excavation
- Capacity and installation sequence of support systems

3. ANALYSIS METHODS

- Spherical projection techniques or analytical methods are used for determination and visualization of all potential wedges in the rock mass surrounding the tunnel
- Limit equilibrium analysis of critical wedges are used for parametric studies on the mode of failure, factor of safety and support requirements.

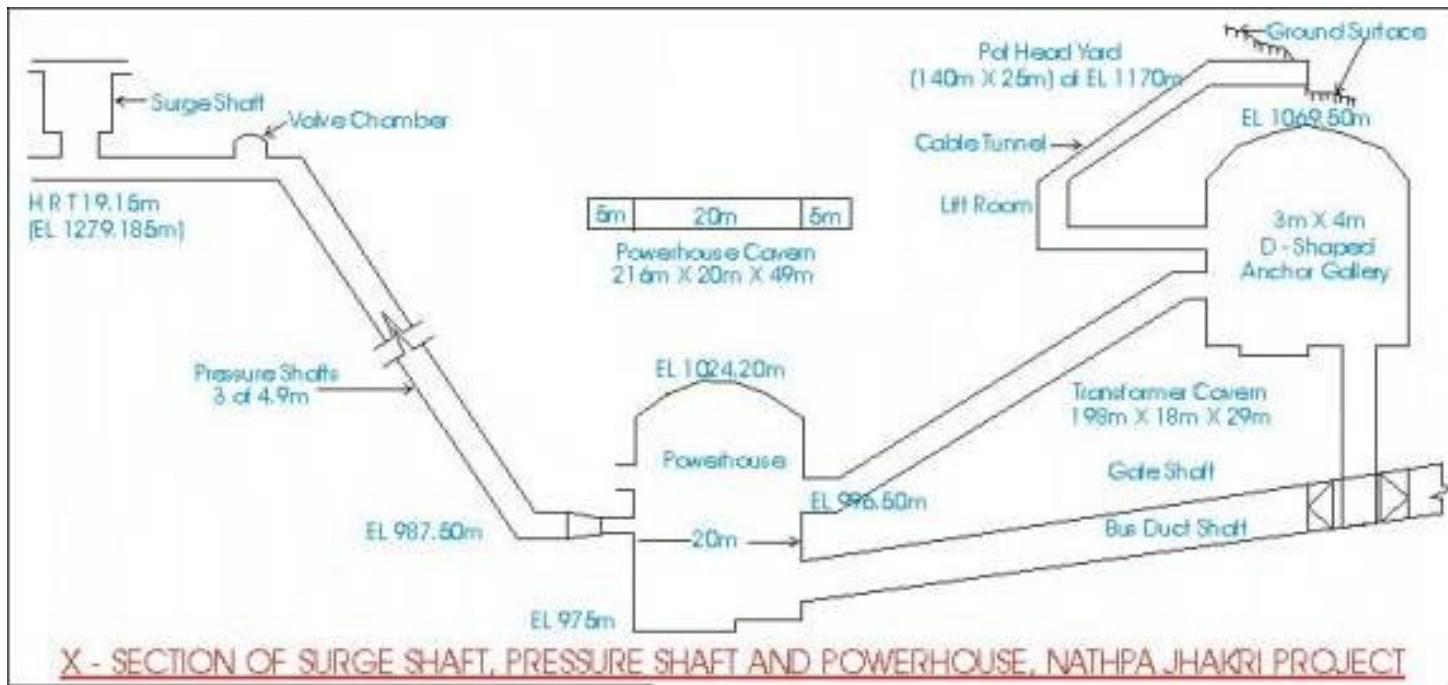


SHALLOW TUNNELS IN JOINTED ROCK



4. ACCEPTIBILITY CRITERIA

- Factor of safety including the effect of reinforcement should exceed 1.5 for sliding and 2.0 for falling wedges and blocks.
- Support installation sequence is critical and wedges or blocks should be identified and supported before they are fully exposed by excavation
- Displacement monitoring is of little value.



LARGE CAVERNS IN JOINTED ROCK



1. TYPICAL PROBLEMS

- Gravity driven falling or sliding wedges or tensile and shear failure of rock mass depending upon spacing of structural feature and magnitude of in situ stresses

2. CRITICAL PARAMETERS

- Shape and orientation of cavern in relation to orientation, inclination and shear strength of structural features in rock mass.
- In situ stresses in rock mass
- Excavation and support sequence and quality of drilling and blasting

3. ANALYSIS METHODS

- Spherical projection techniques or analytical methods are used for determination and visualization of all potential wedges in the rock mass surrounding the cavern
- Stress and displacement induced by each stage of cavern excavation are determined by numerical analyses and are used to estimate support requirements for the cavern roof and walls.

4. ACCEPTIBILITY CRITERIA

- An acceptable design is achieved when numerical model indicates that the extent of failure has been controlled by installed support,
- that the support is not overstressed
- That the displacement in the rock mass is stabilized
- Monitoring of displacement is essential to confirm design predictions.

1. TYPICAL PROBLEMS

- Stresses and/or thermally induced spalling of the rock surrounding the excavation.
- This results in increased permeability and higher probability of radioactive leakage

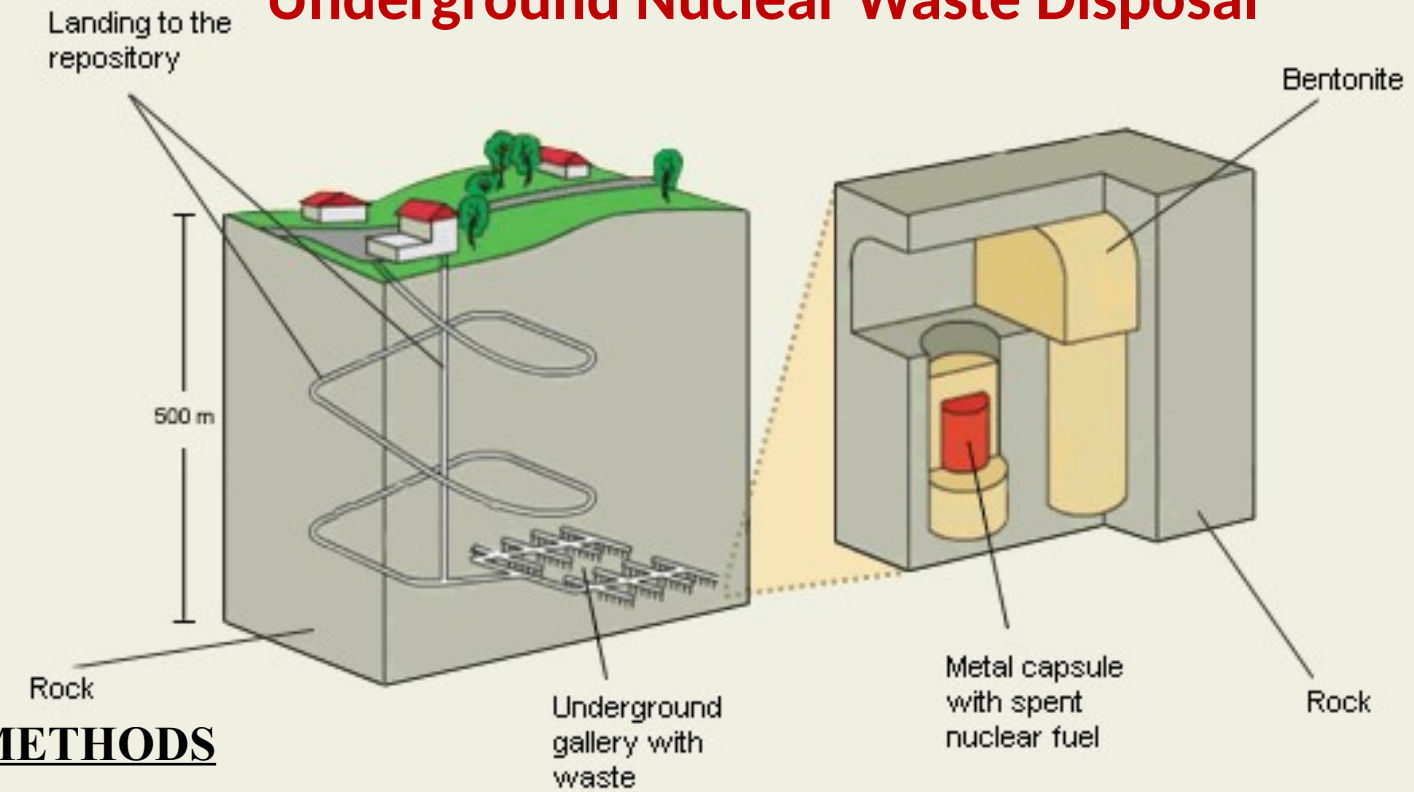
2. CRITICAL PARAMETERS

- Orientation, inclination, permeability and shear strength of structural features in rock mass
- In situ and thermal stresses in the rock surrounding the excavations
- Groundwater distribution in the rock mass.

3. ANALYSIS METHODS

- Numerical analyses are used to calculate stresses and displacements induced by excavation and by thermal loading from waste canisters
- Ground water flow pattern and velocities, particularly through blast damaged zones, fissures in rock and shaft seals are calculated using numerical methods.

Underground Nuclear Waste Disposal



4. ACCEPTIBILITY CRITERIA

- An acceptable design requires extremely low rates of ground water movement through the waste containment area in order to limit transport of radioactive material.
- Shafts, tunnels and canister holes must remain stable for approximately 50 years to permit retrieval of waste if necessary.

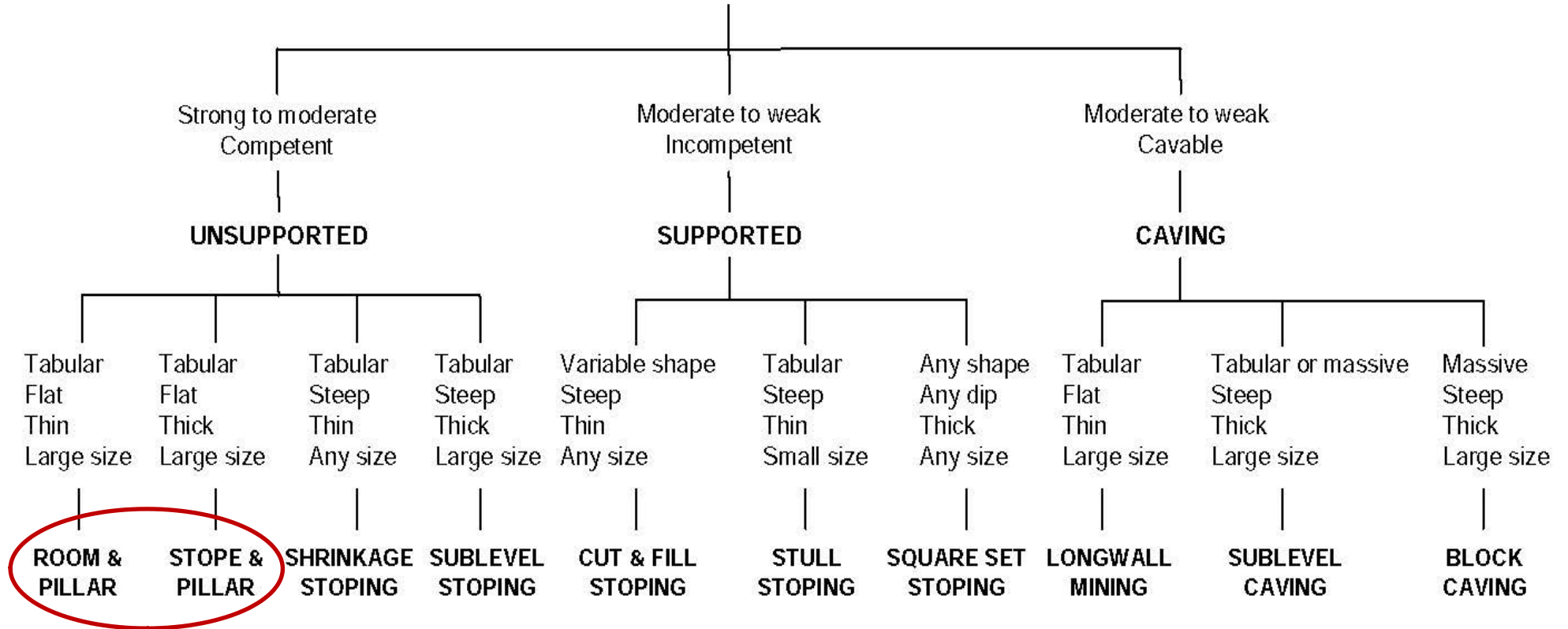
UNDERGROUND MINING METHODS

Ore strength:
Rock strength:

Stoping class:

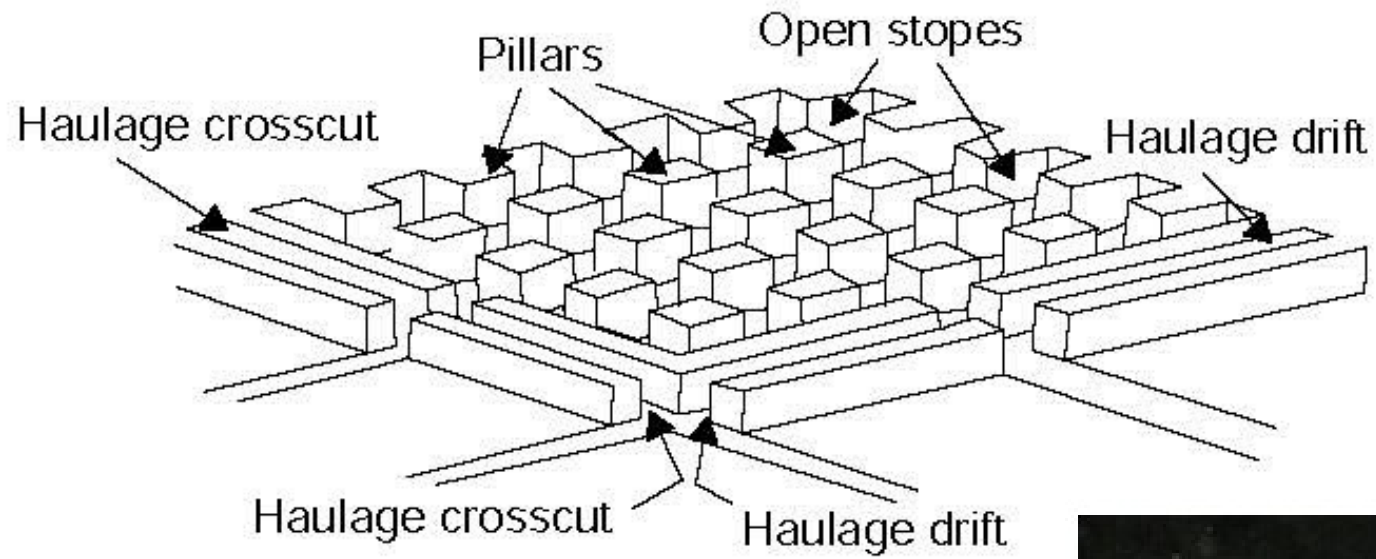
Deposits' geometry:

Stoping method:



Pillar Design

Room and Pillar Mining



1. TYPICAL PROBLEMS

- Progressive spalling and slabbing of rock mass leading to eventual pillar collapse or rock bursting

Room and Pillar Mining

2. CRITICAL PARAMETERS

- Strength of rock mass forming the pillars
- Presence of unfavourably oriented structural features
- Pillar geometry – (W/H – ratio)
- Overall mine geometry including extraction ratio.

3. ANALYSIS METHODS

- For horizontally bedded deposits, pillar strength from empirical relationships based on W/H ratio and average pillar stress based on tributary area calculation compared to give FoS.
- For more complex mining geometry, numerical analysis including progressive pillar failure may be required.

4. ACCEPTIBILITY CRITERIA

- FoS for simple pillar layout in horizontally bedded deposit should exceed 1.5 for “Premanent “ pillars
- Incase where progressive failure of complex pillar layout is modelled, individual pillar failure can be tolerated provided that they do not initiate “domino” failure of adjacent pillars.

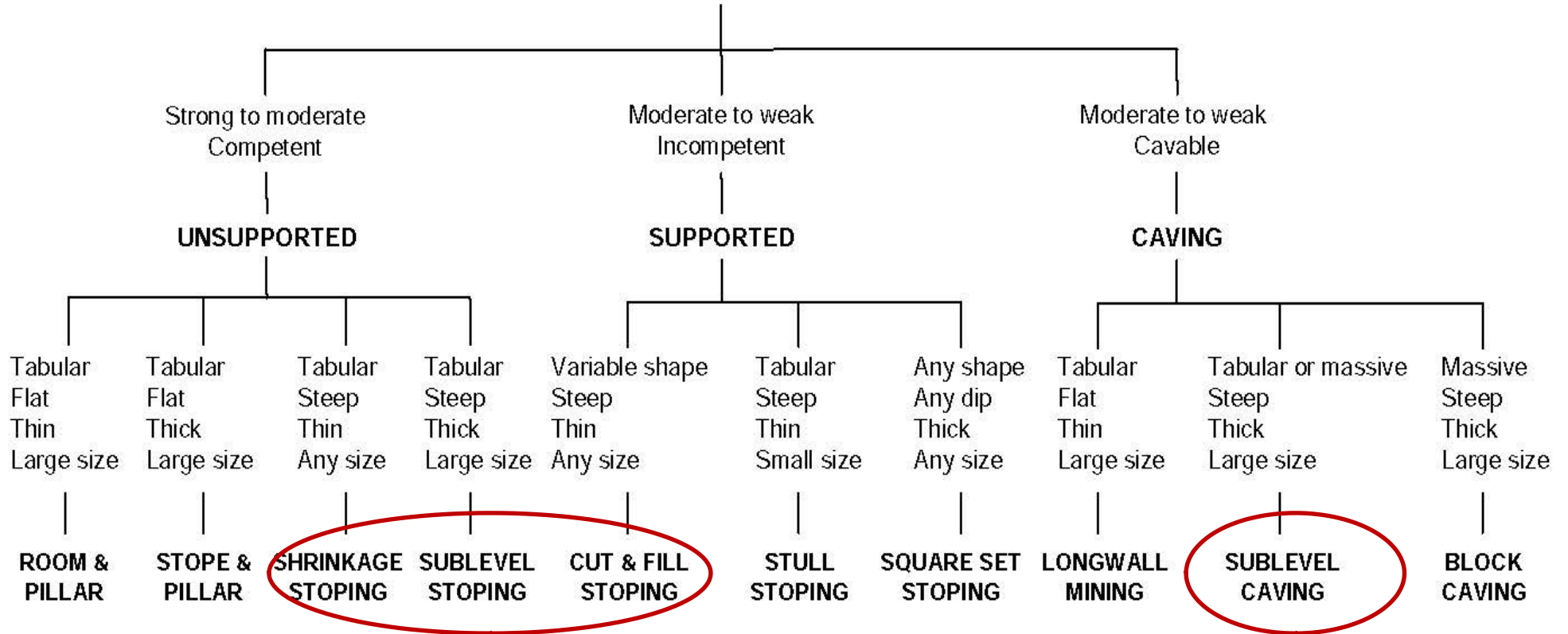
UNDERGROUND MINING METHODS

Ore strength:
Rock strength:

Stoping class:

Deposits' geometry:

Stoping method:



Crown Pillars

Crown Pillars in U/g Metal Mines

1. TYPICAL PROBLEMS

- Caving of surface crown pillars for which the ratio of pillar depth to stope span is inadequate.
- Rock bursting or gradual spalling of overstressed internal crown pillars.

2. CRITICAL PARAMETERS

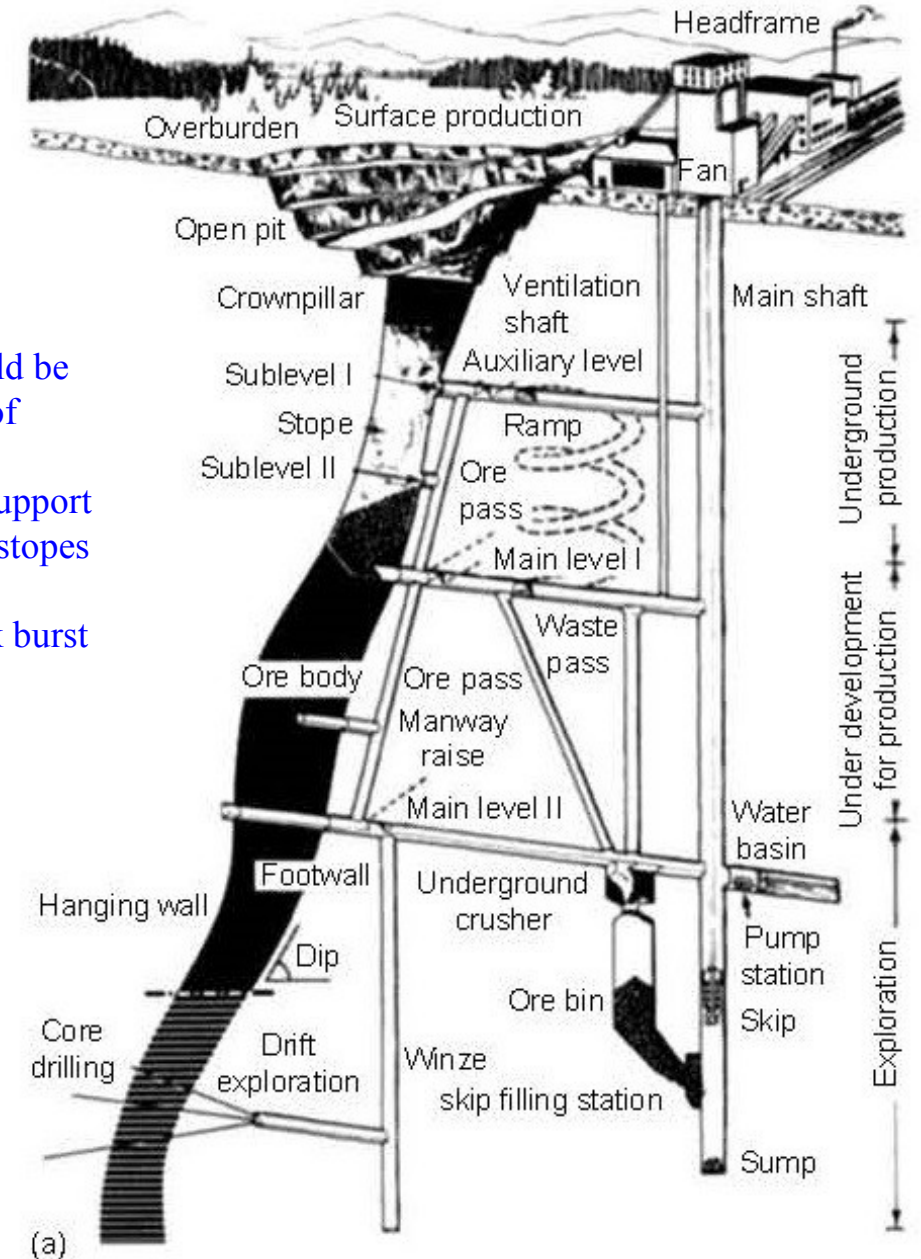
- Strength of rock mass forming the pillars
- Depth of weathering and presence of steeply dipping structural feature in case of surface crown pillars
- In situ stress levels and geometry of internal crown pillars

3. ANALYSIS METHODS

- Rock mass classification and limit equilibrium analysis can give useful guidance on surface crown pillar dimensions for different rock masses
- Numerical analyses, including discrete element simulations, can give approximate stress levels and indications of zones of potential failure.

4. ACCEPTIBILITY CRITERIA

- Surface crown pillar depth to span ratio should be large enough to ensure very low probability of failure
- Internal crown pillar may require extensive support to ensure stability during mining of adjacent stopes
- Careful planning of mining sequence may be necessary to avoid high stress levels and rock burst problems.



UNDERGROUND MINING METHODS

Ore strength:
Rock strength:

Stoping class:

Deposits' geometry:

Stoping method:

Strong to moderate
Competent

Moderate to weak
Incompetent

Moderate to weak
Cavable

UNSUPPORTED

SUPPORTED

CAVING

Tabular
Flat
Thin
Large size

Tabular
Flat
Thick
Large size

Tabular
Steep
Thin
Any size

Tabular
Steep
Thick
Large size

Variable shape
Steep
Thin
Any size

Tabular
Steep
Thin
Small size

Any shape
Any dip
Thick
Any size

Tabular
Flat
Thin
Large size

Tabular or massive
Steep
Thick
Large size

Massive
Steep
Thick
Large size

**ROOM &
PILLAR**

**STOPE &
PILLAR**

**SHRINKAGE
STOPING**

**SUBLEVEL
STOPING**

**CUT & FILL
STOPING**

**STULL
STOPING**

**SQUARE SET
STOPING**

**LONGWALL
MINING**

**SUBLEVEL
CAVING**

**BLOCK
CAVING**

1. TYPICAL PROBLEMS

- Falls of structurally defined wedges and blocks from stope backs and hanging walls
- Stress induced failures and rockbursting in high stress environments

2. CRITICAL PARAMETERS

- Orientation, inclination and shear strength of structural features in rock mass
- In situ stresses in rock mass
- Shape and orientation of stope
- Quality, placement and drainage of fill

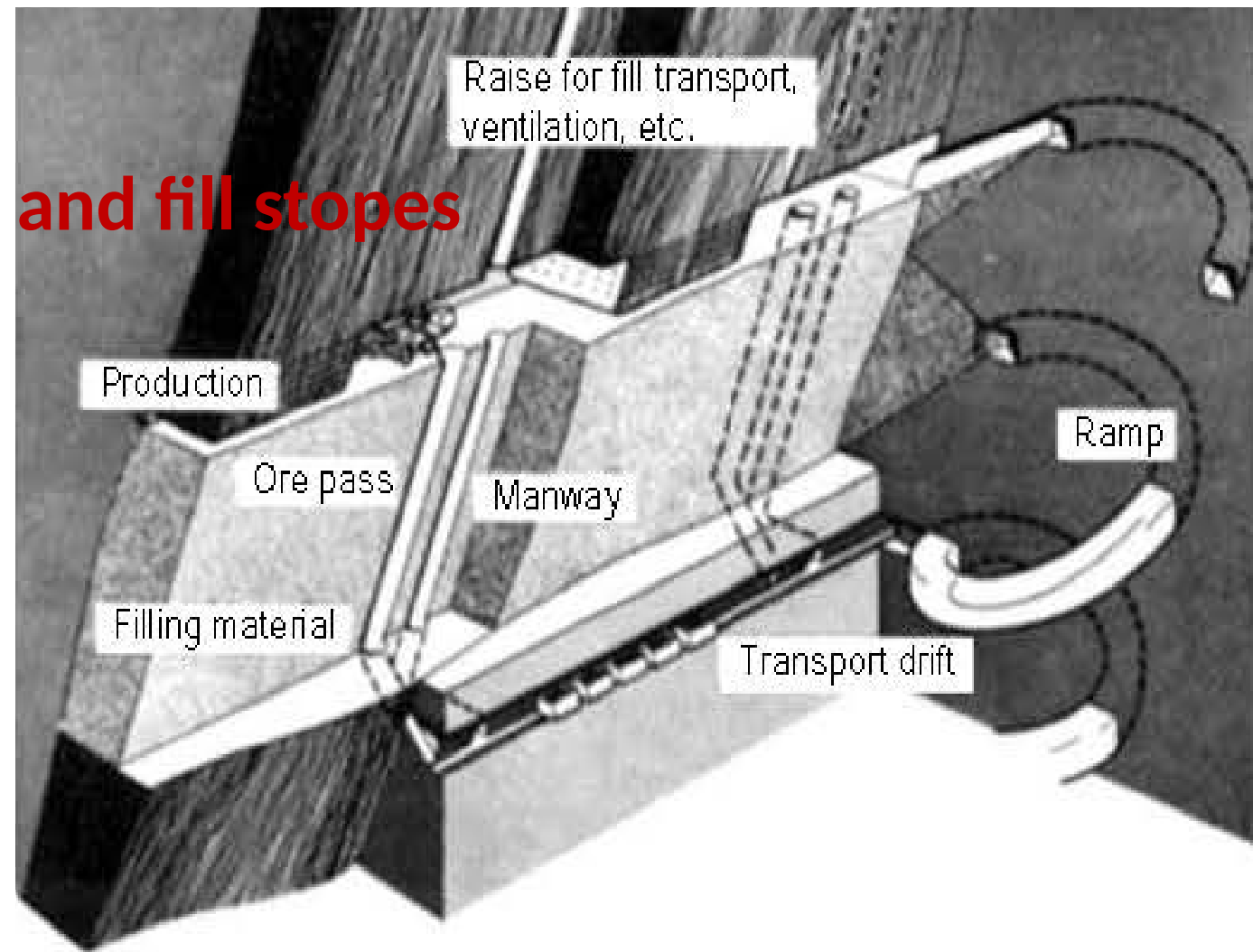
3. ANALYSIS METHODS

- Numerical analyses of stresses and displacements for each excavation stage will give some indication of potential problems
- Some of more sophisticated numerical models will permit inclusion of the support provided by fill or the reinforcement of the rock by means of grouted cables

4. ACCEPTIBILITY CRITERIA

- Local instability should be controlled by the installation of the rockbolts or grouted cables to improve safety and to minimize ore dilution
- Overall stability is controlled by the geometry and excavation sequence of the stopes and the quality and sequence of filling
- Acceptable mining conditions are achieved when all the ore is recovered safely.

Cut and fill stopes



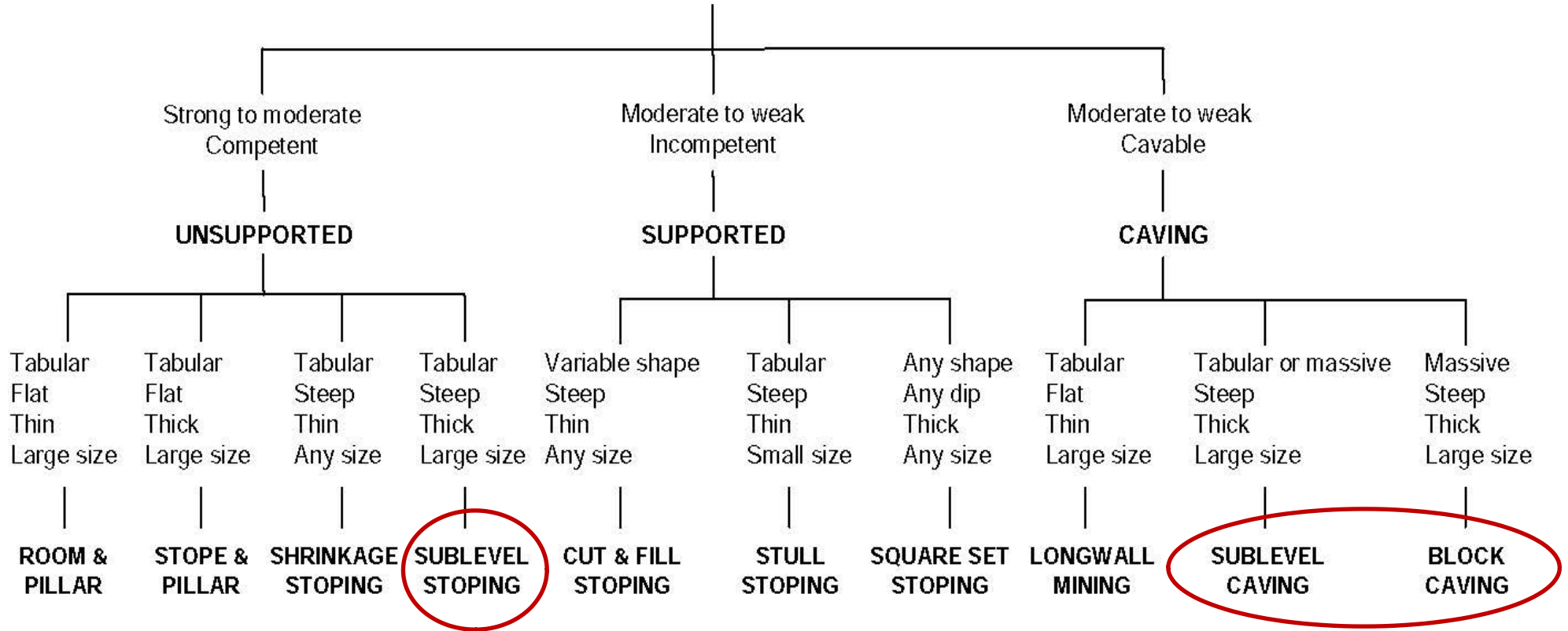
UNDERGROUND MINING METHODS

Ore strength:
Rock strength:

Stope class:

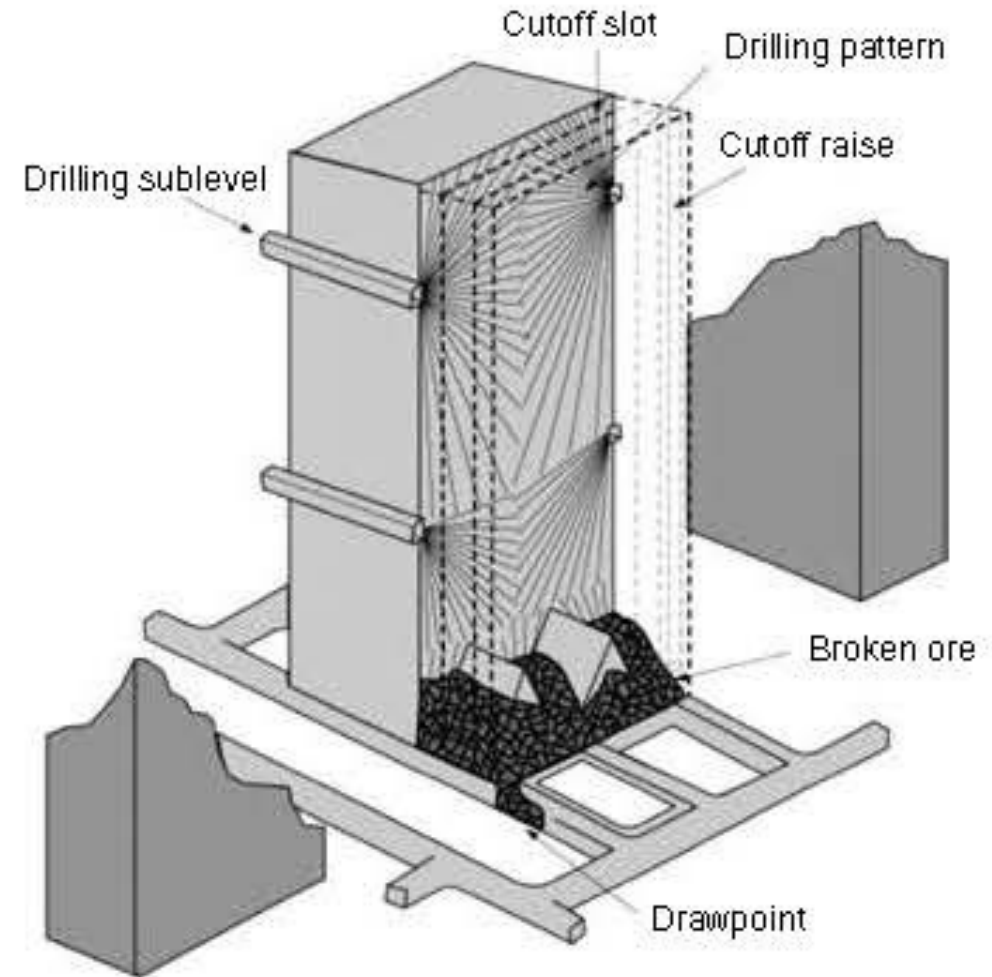
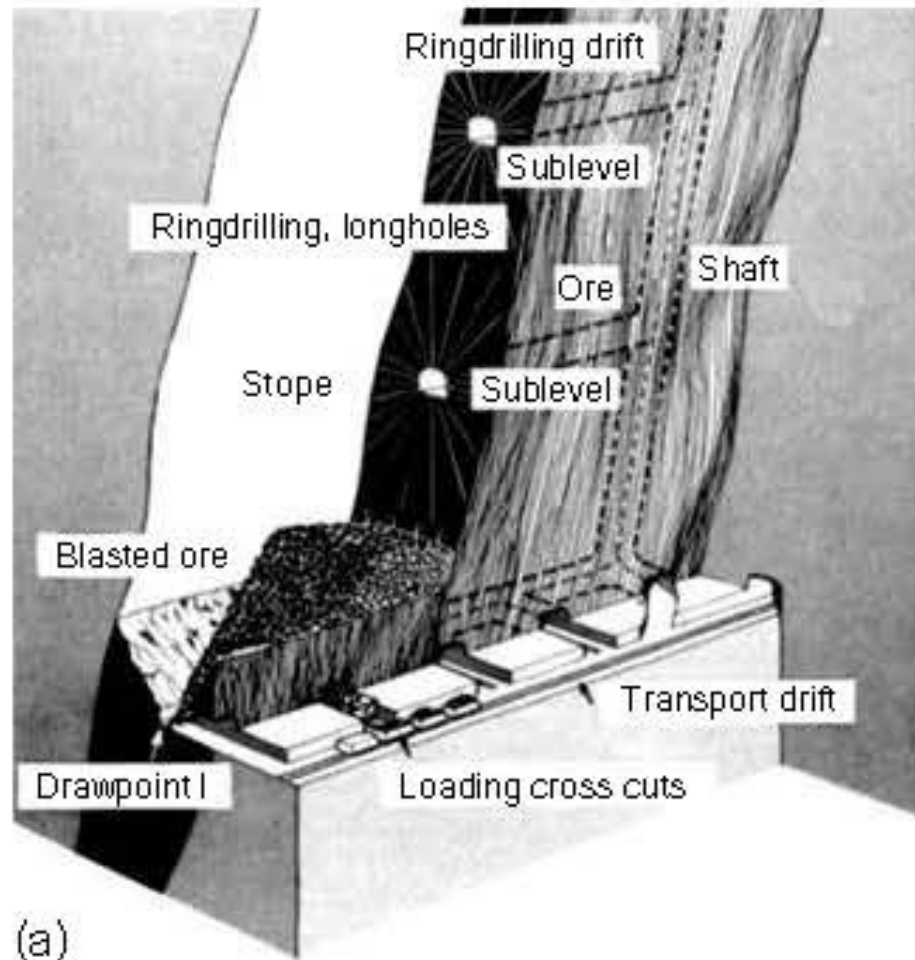
Deposits' geometry:

Stope method:



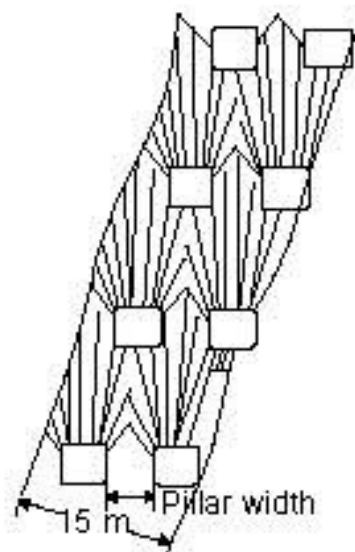
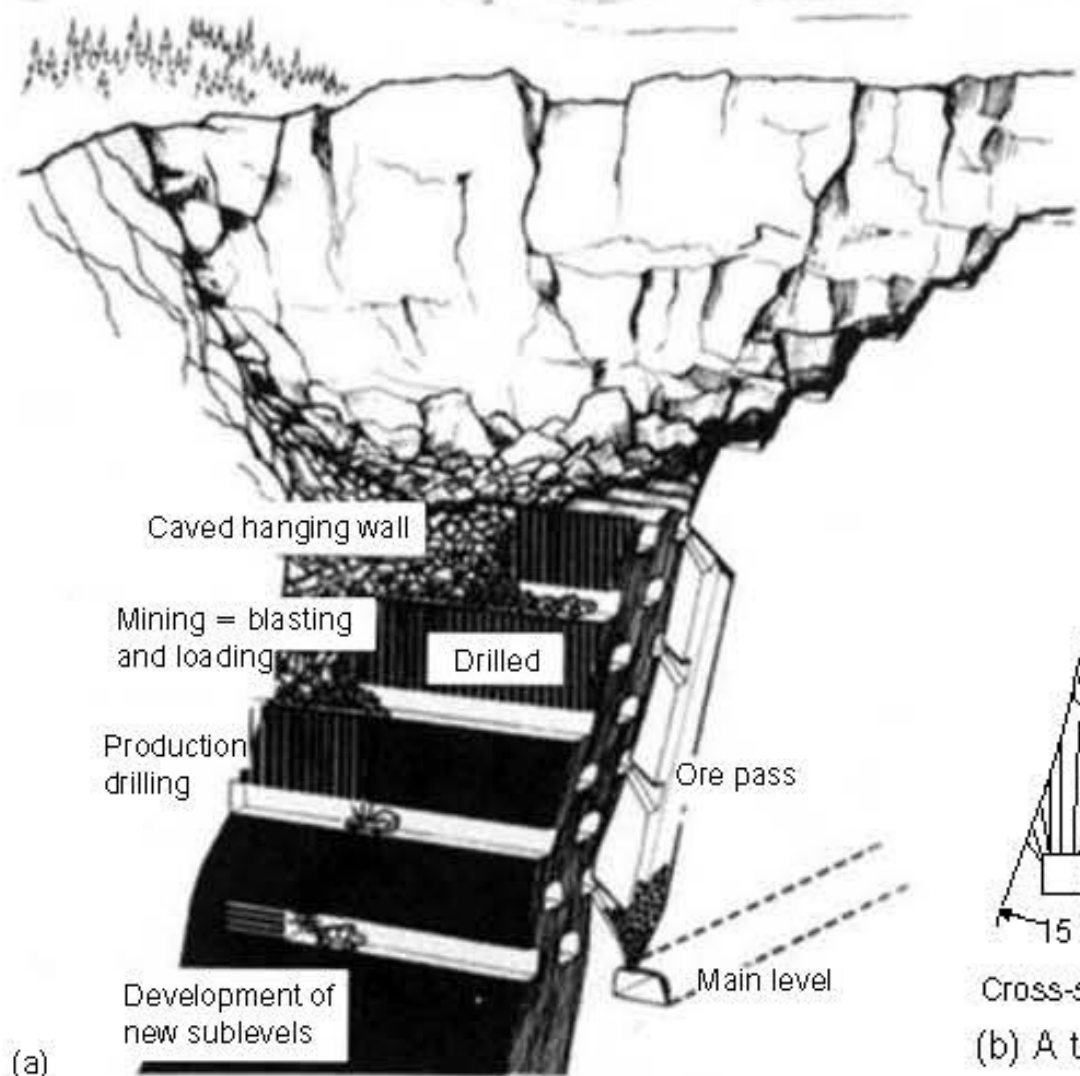
Non-entry stopes

Sublevel stopping:- Non-entry stope

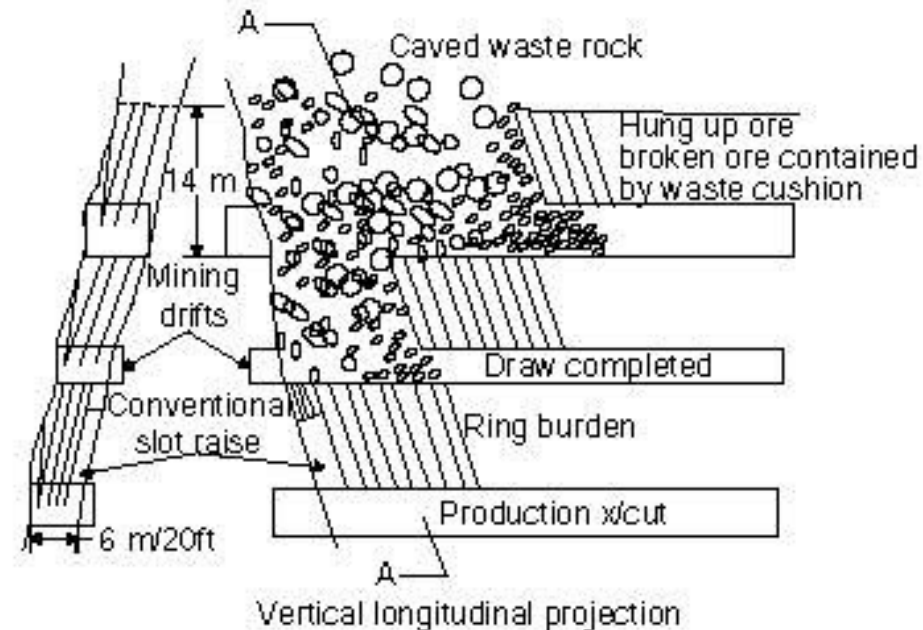


Sublevel stopping Mount Isa

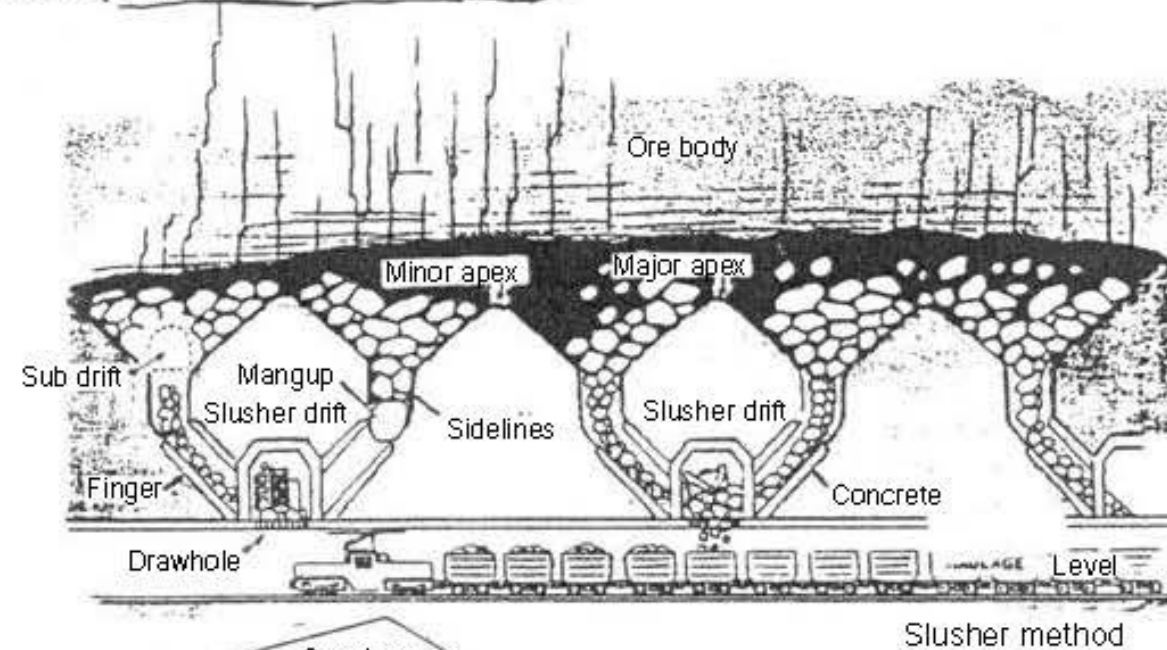
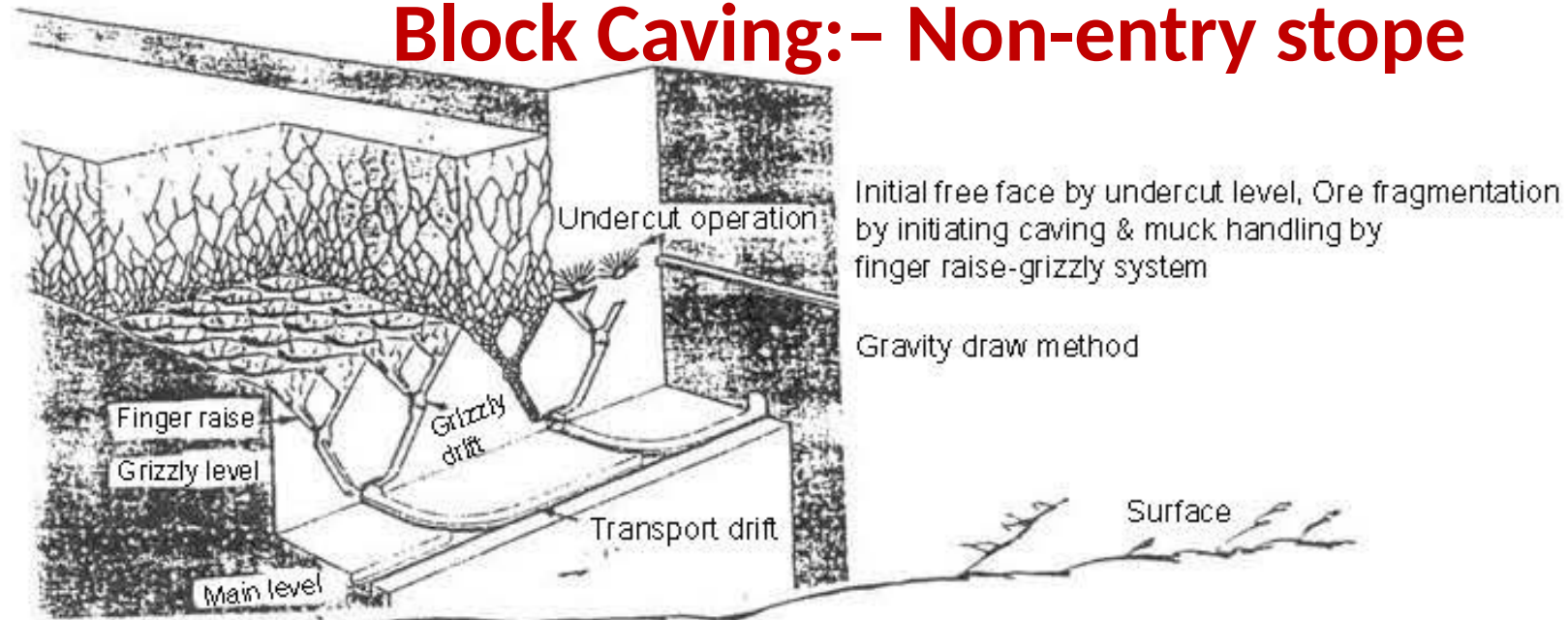
Sublevel Caving:- Non-entry stope



(b) A typical longitudinal sublevel-caving stope



Block Caving:- Non-entry stope



Non-entry stopes

1. TYPICAL PROBLEMS

- Ore dilution resulting from rockfalls from stope back and walls
- Rockbursting or progressive failure induced by high stresses in pillars between stopes

2. CRITICAL PARAMETERS

- Quality and strength of rock
- In situ and induced stresses in the rock surrounding the excavations
- Quality of drilling and blasting in excavation of the stope

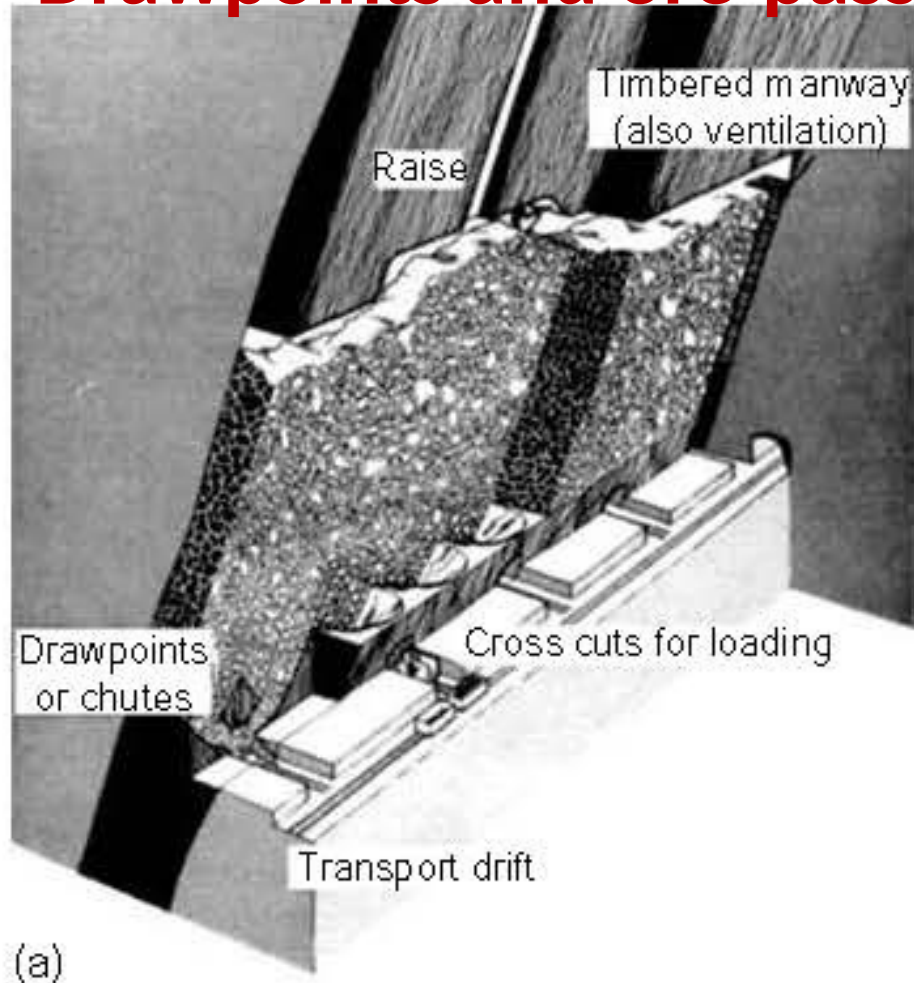
3. ANALYSIS METHODS

- Some empirical rules based on rock mass classification are available for estimating safe stope dimensions
- Numerical analyses of stope layout and mining sequence, using three dimensional analyses for complex orebody shapes will provide indications of potential problems and estimates of support requirements

4. ACCEPTIBILITY CRITERIA

- A design of this type can be considered acceptable when safe and low cost recovery of large proportion of orebody has been achieved.
- Rockfalls in shafts and haulages are an unacceptable safety hazard and a pattern of support may be required
- In high stress environments, local distressing may be used to reduce rockbursting

Drawpoints and ore-passes



1. TYPICAL PROBLEMS

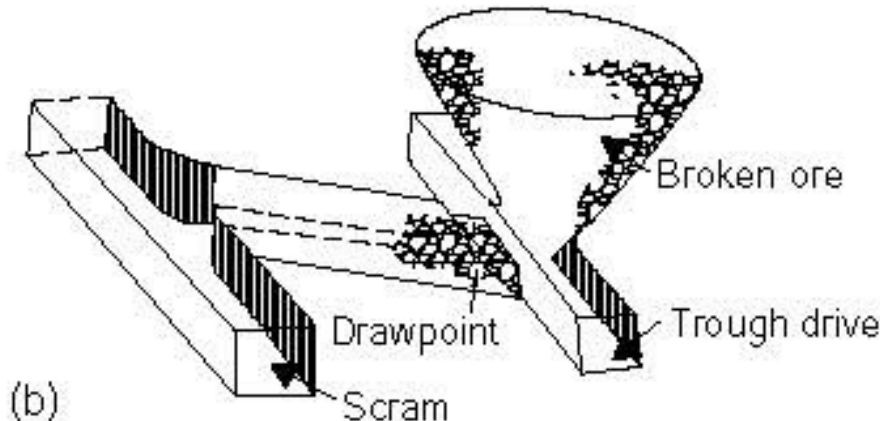
- Local rock mass failure resulting from abrasion and wear of poorly supported drawpoints or orepasses.
- In extremes cases this may result on loss stopes or orepasses.

2. CRITICAL PARAMETERS

- Quality and strength of rock
- In situ and induced stresses and stress changes in rock surrounding the excavations
- Selection and installation sequence of support

3. ANALYSIS METHODS

- Limit equilibrium or numerical analyses are not particularly useful since the wear and abrasion are not included in these models
- Empirical designs based upon previous experience or trial and error methods are generally used.



4. ACCEPTABILITY CRITERIA

- The shape of opening should be maintained for the design of life of the drawpoint or orepass. Loss of control can result in serious dilution of ore or abandonment of the excavation
- Wear resistant flexible reinforcement such as grouted cables, installed during excavation of the opening may be successful in controlling instability.